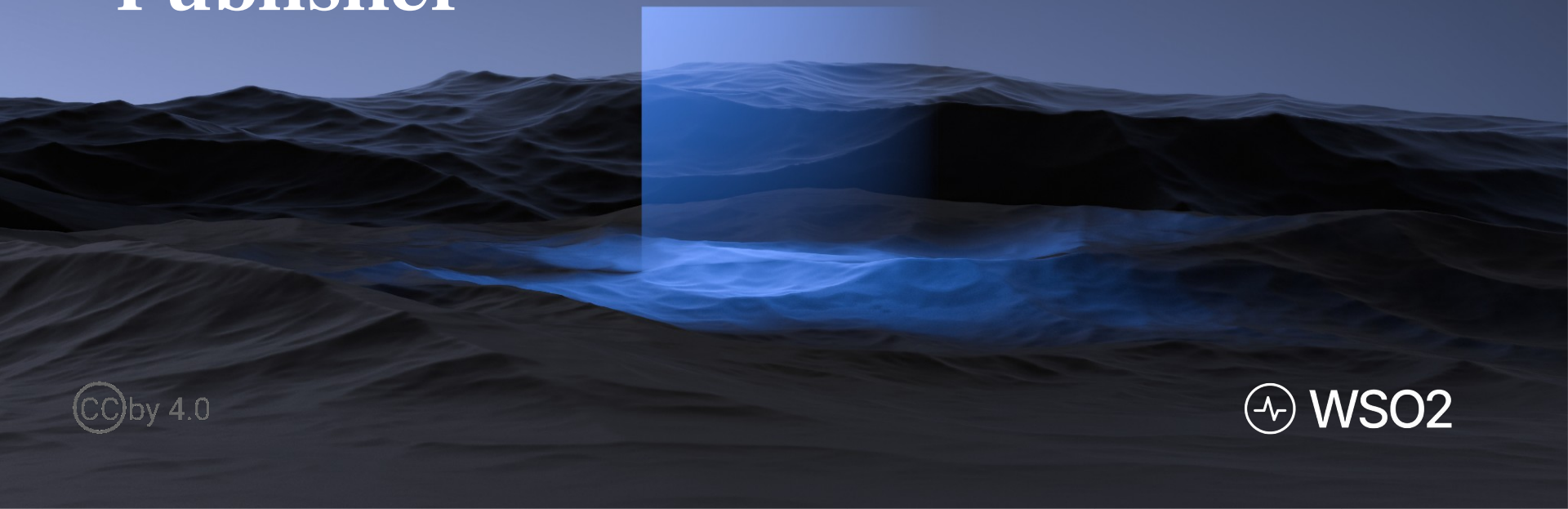
Getting Started with the API Publisher



CC by 4.0

WSO2

What is the API Publisher Portal?

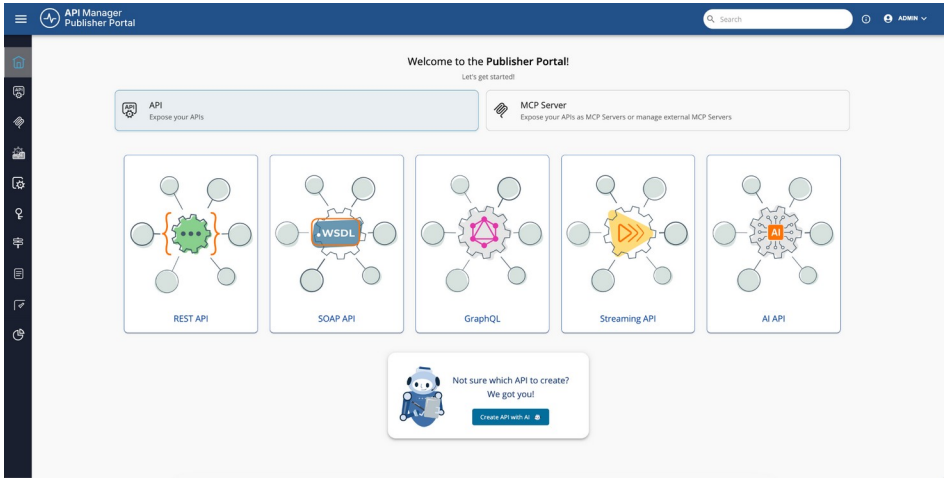
- React Web Application
- Story of APIs start here.
- Objectives
    - Design APIs
    - Deploy APIs
    - Publish APIs
    - Manage APIs

Creating an API

WSO2

Ways of Creating APIs

- Design a new REST API
- REST API from OpenAPI or Swagger definition
- REST API from SOAP backend
- SOAP API as a REST service
- GraphQL API from SDL
- Design new WebSocket API
- Design Webhook/WebSub API
- Create SSE (Server-Sent Events) API
- Import an Async API
- Import from Service Catalog
- Add third party APIs
- AI/LLM API



Creating APIs

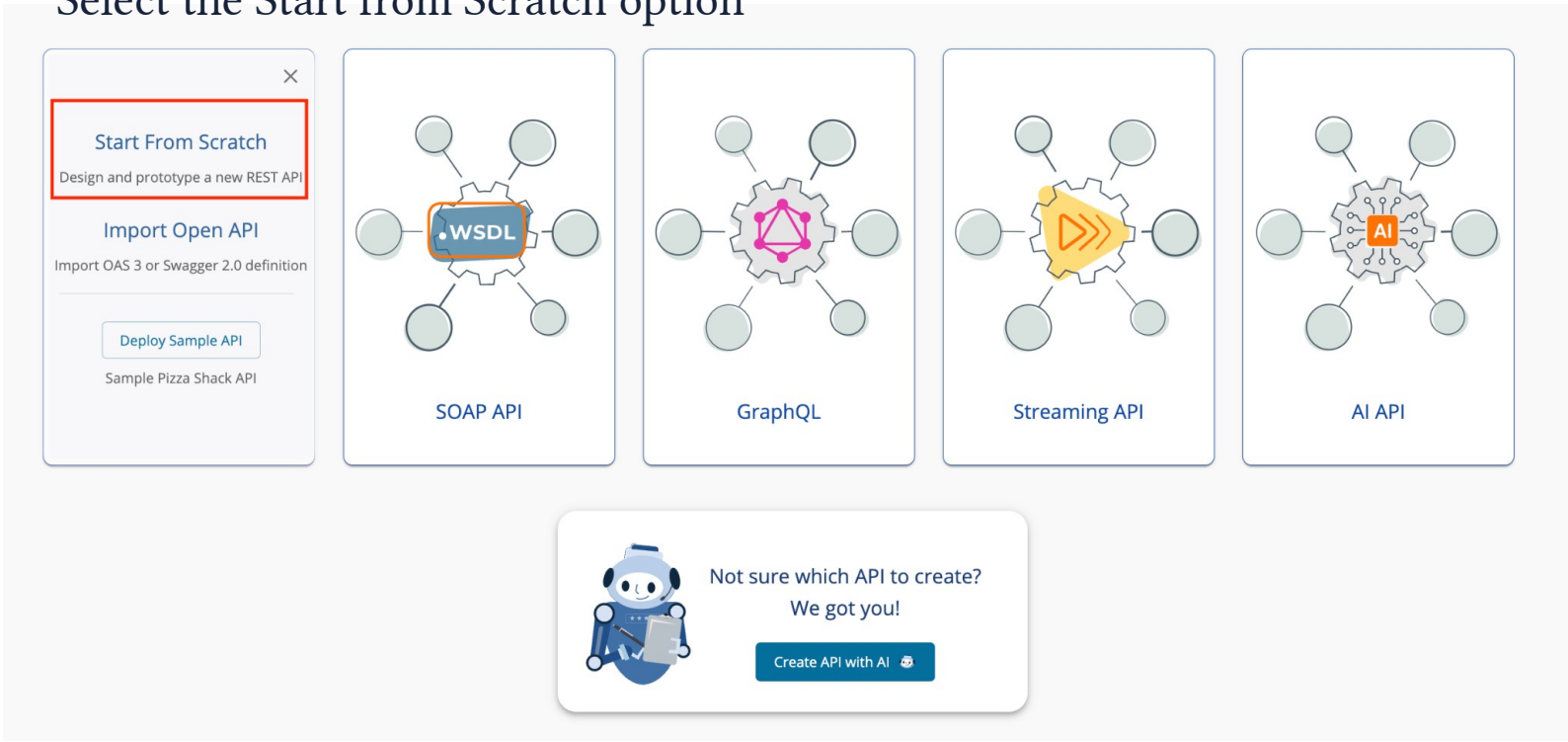
Let’s try it out!

Creating and publishing an API through the Publisher

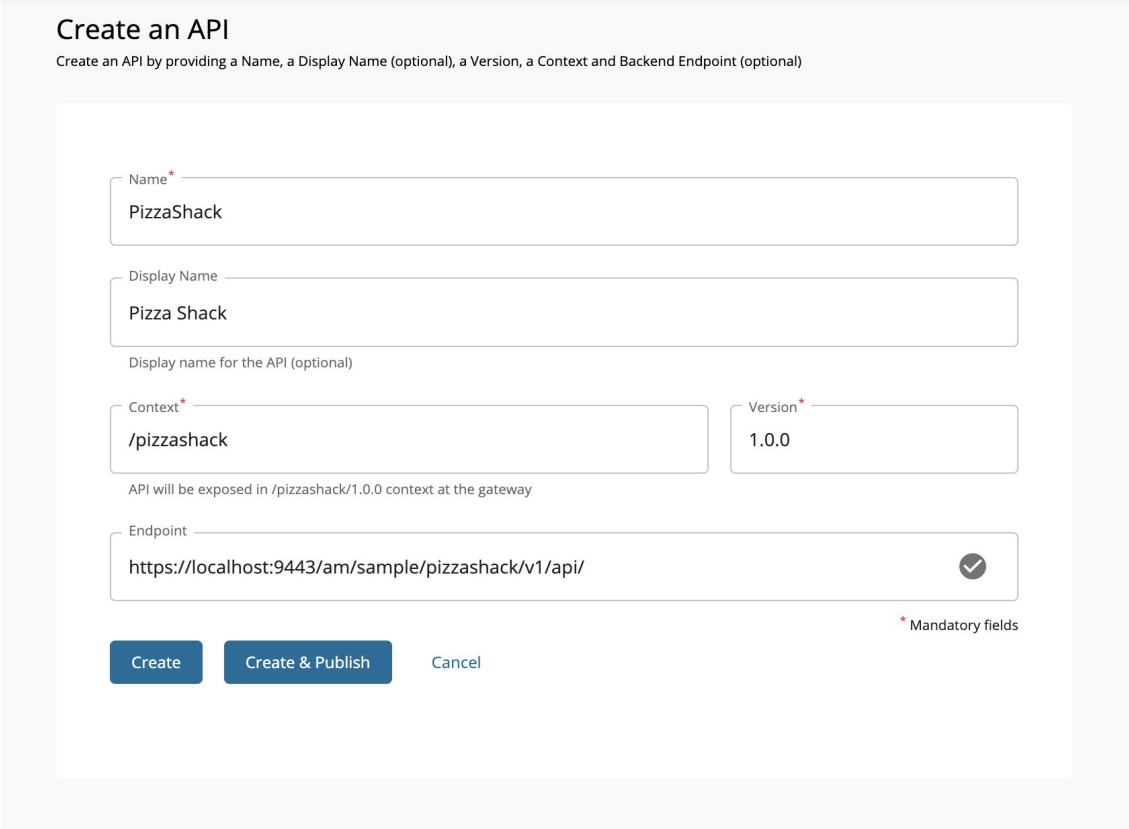
WSO2

Design a New REST API

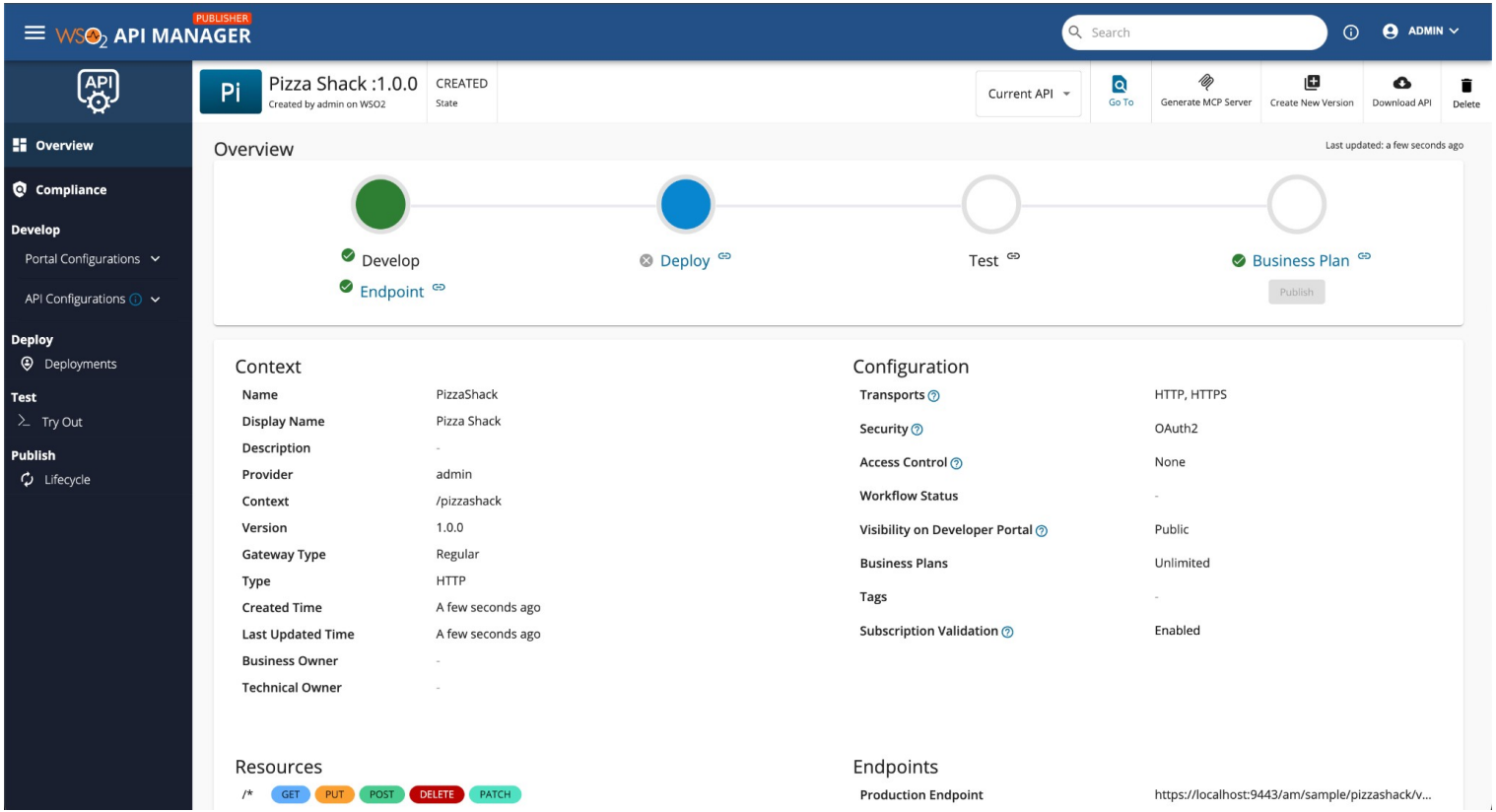
Select the Start from Scratch option



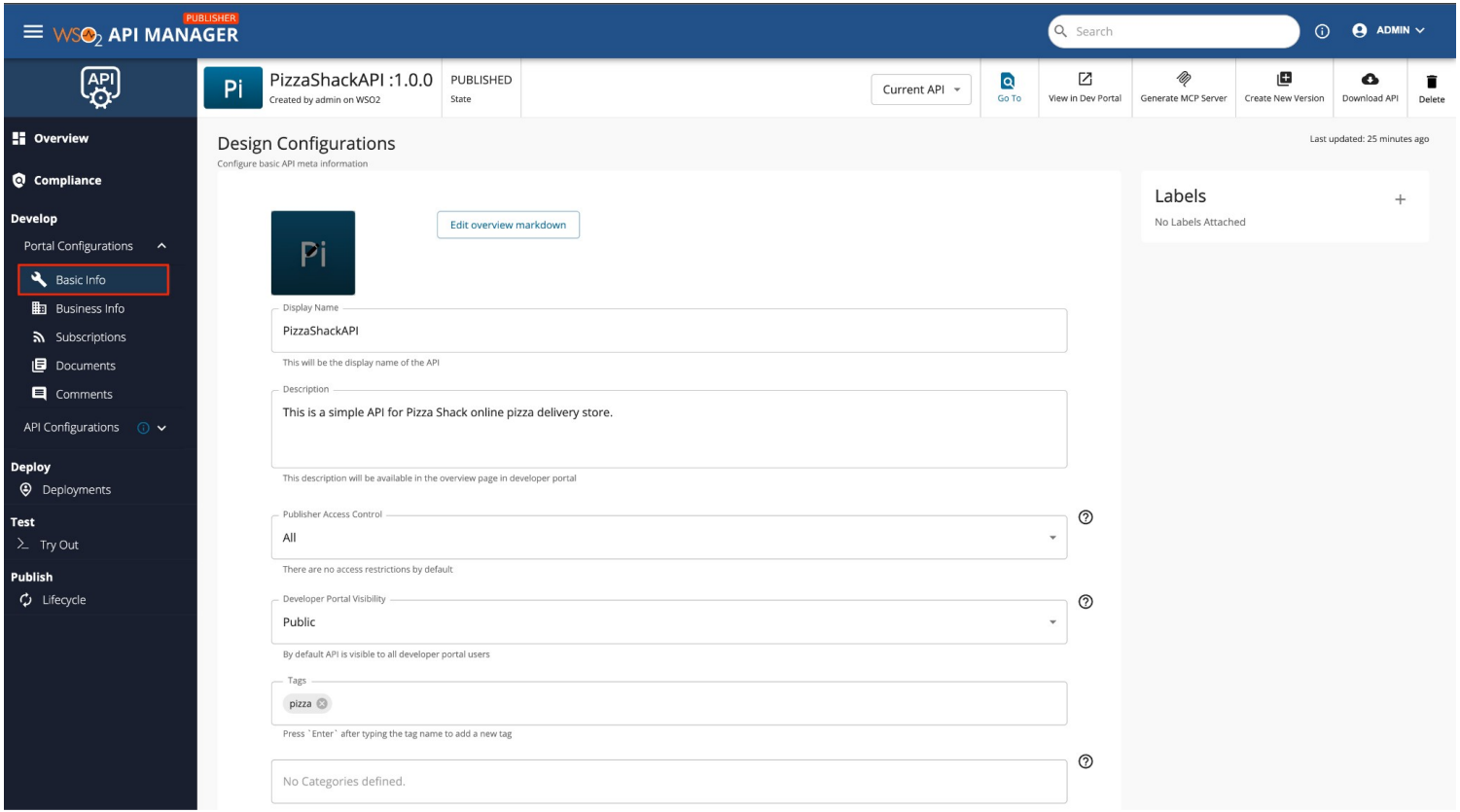
Basic Details



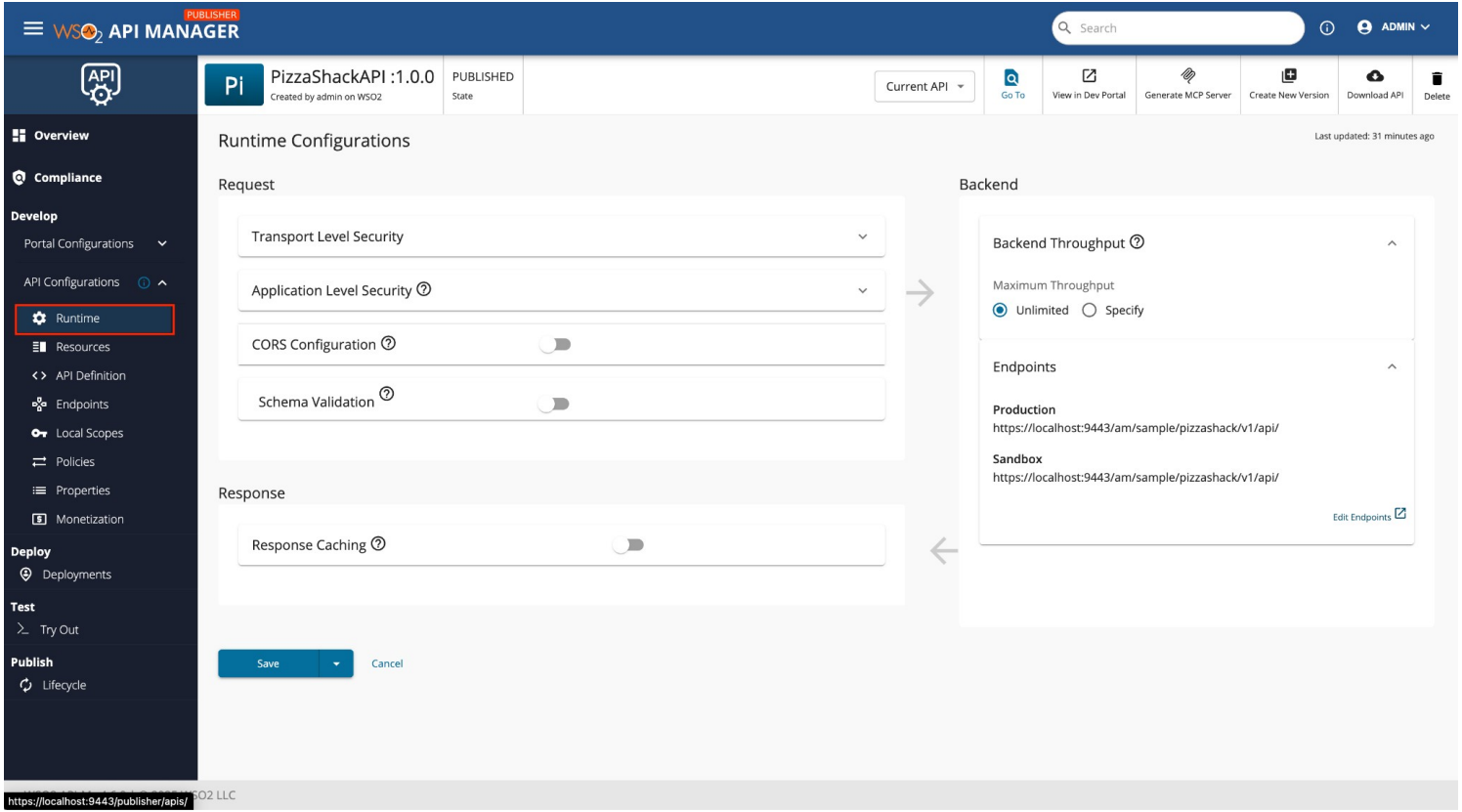
API Overview



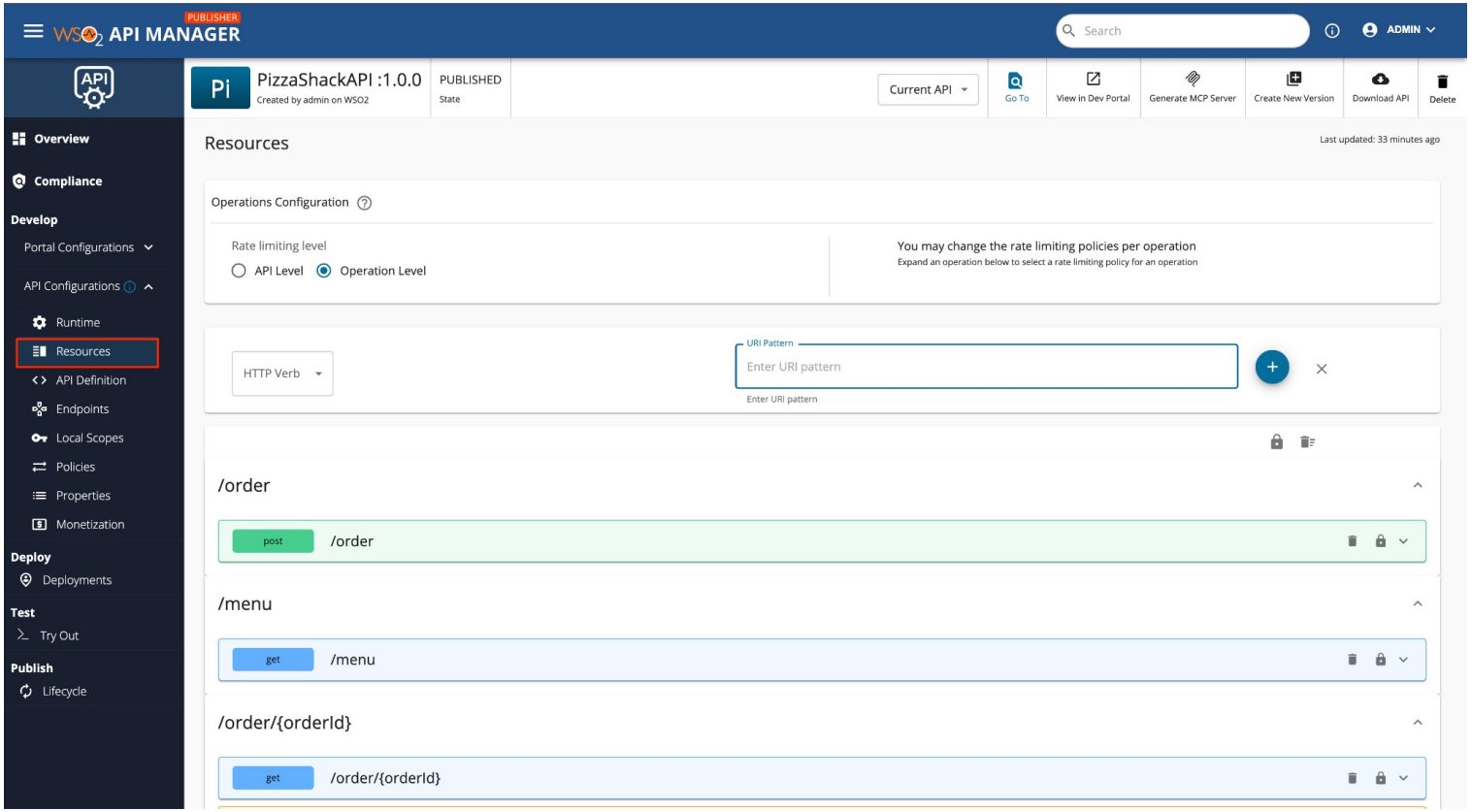
API Overview



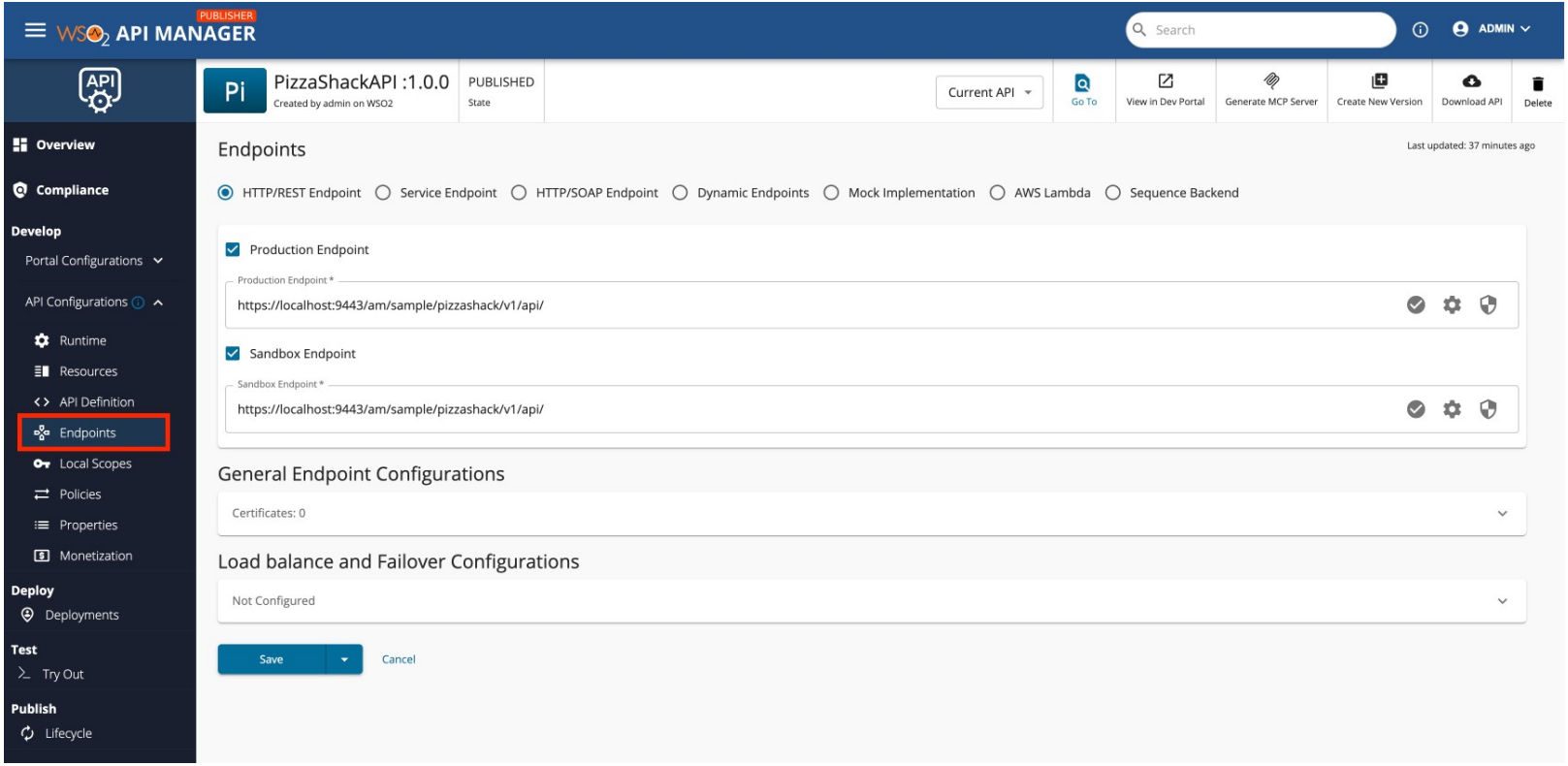
API Runtime Configurations



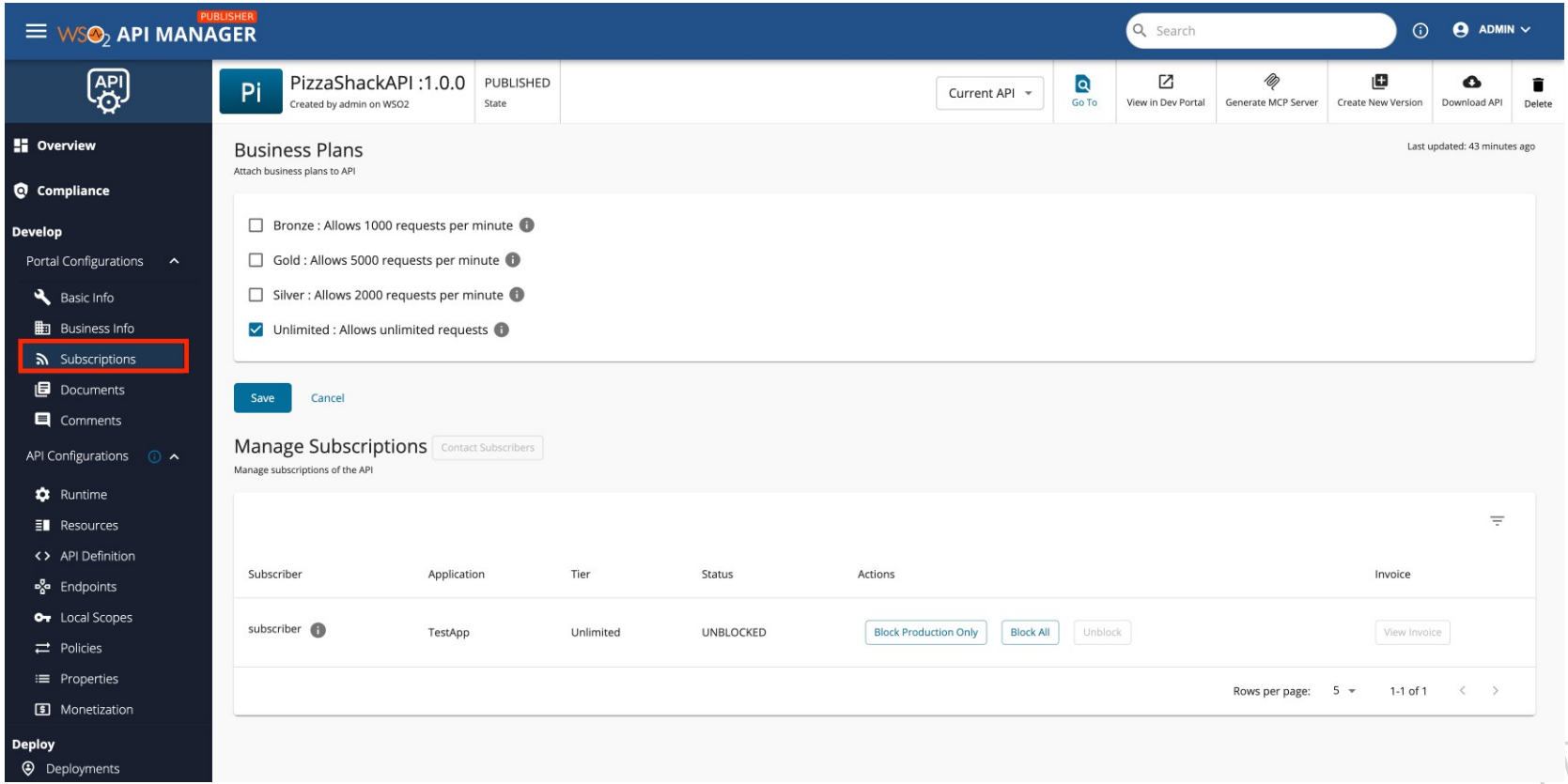
API Resources



Endpoints


PUBLISHER
WSO2 API MANAGER
Search
ADMIN
API
Pi
PizzaShackAPI :1.0.0
Created by admin on WSO2
PUBLISHED
State
Current API
Go To
View in Dev Portal
Generate MCP Server
Create New Version
Download API
Delete
Overview
Compliance
Develop
Portal Configurations
API Configurations
Runtime
Resources
API Definition
Endpoints
Local Scopes
Policies
Properties
Monetization
Deploy
Deployments
Test
Try Out
Publish
Lifecycle
Endpoints
Last updated: 37 minutes ago
HTTP/REST Endpoint
Service Endpoint
HTTP/SOAP Endpoint
Dynamic Endpoints
Mock Implementation
AWS Lambda
Sequence Backend
Production Endpoint
Production Endpoint *
https://localhost:9443/am/sample/pizzashack/v1/api/
Sandbox Endpoint
Sandbox Endpoint *
https://localhost:9443/am/sample/pizzashack/v1/api/
General Endpoint Configurations
Certificates: 0
Load balance and Failover Configurations
Not Configured
Save
Cancel

Business Plans and Subscriptions

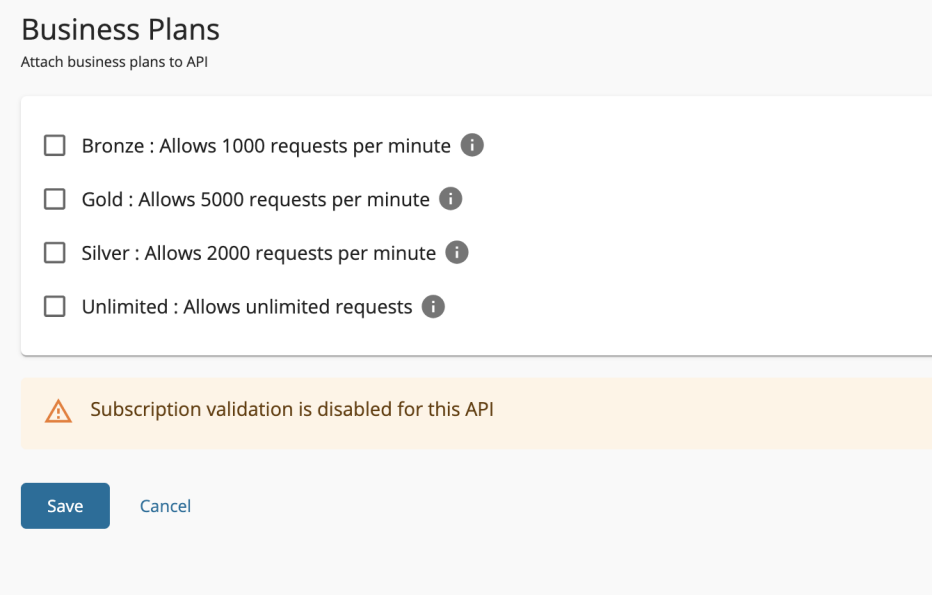


Disabling Subscription Requirement for APIs

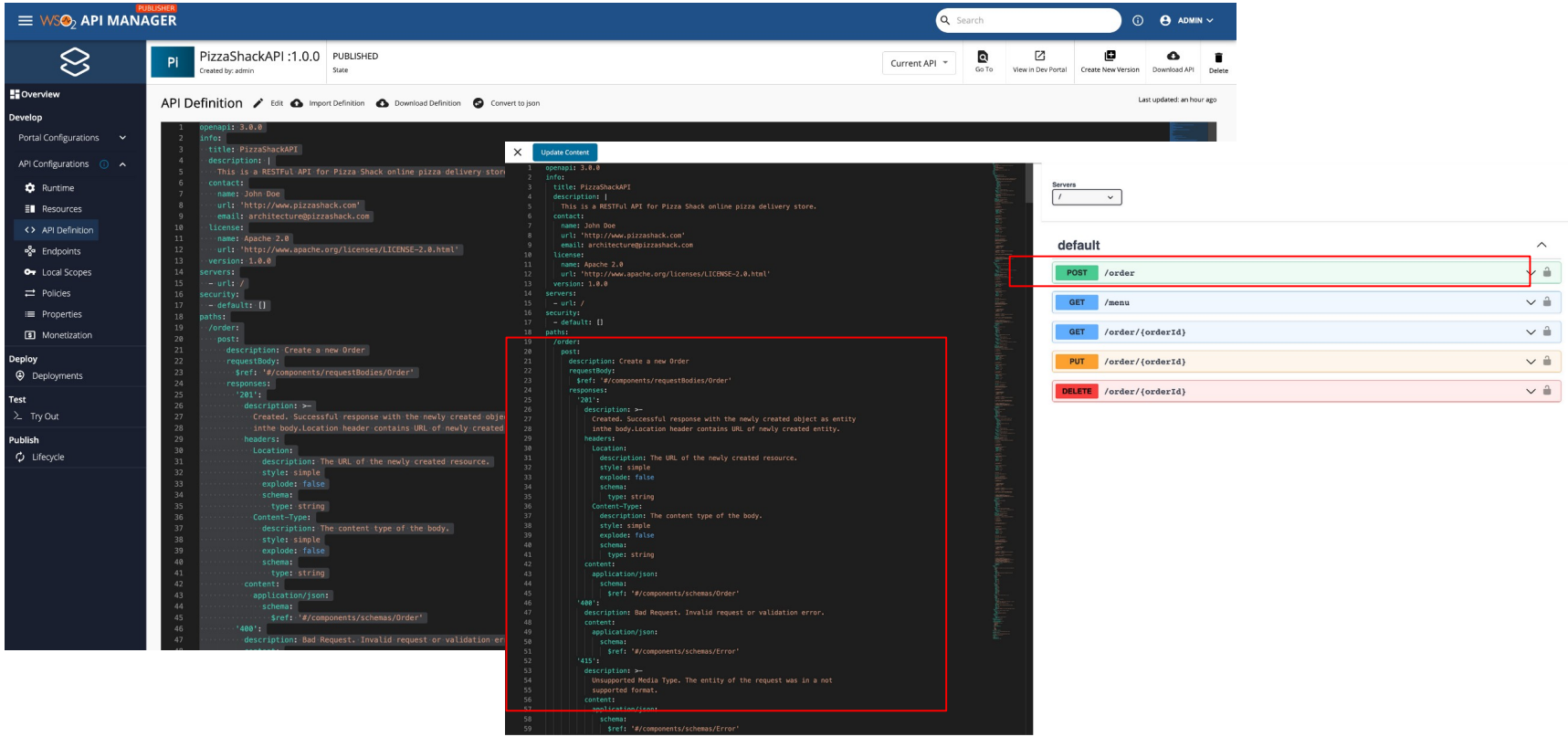
- By default subscriptions are mandatory for API consumption in WSO2 API Manager.
- But, there could be scenarios where an API Publisher might want to publish an API that does not require subscriptions for API consumption.
- Subscription requirement can be disabled for an API at any point after the creation of the API. But it is recommended to do so before publishing the API.
- Once the subscription requirement has been disabled, the gateway will not validate the subscription when an API request comes to the gateway.

Disabling Subscription Requirement for APIs

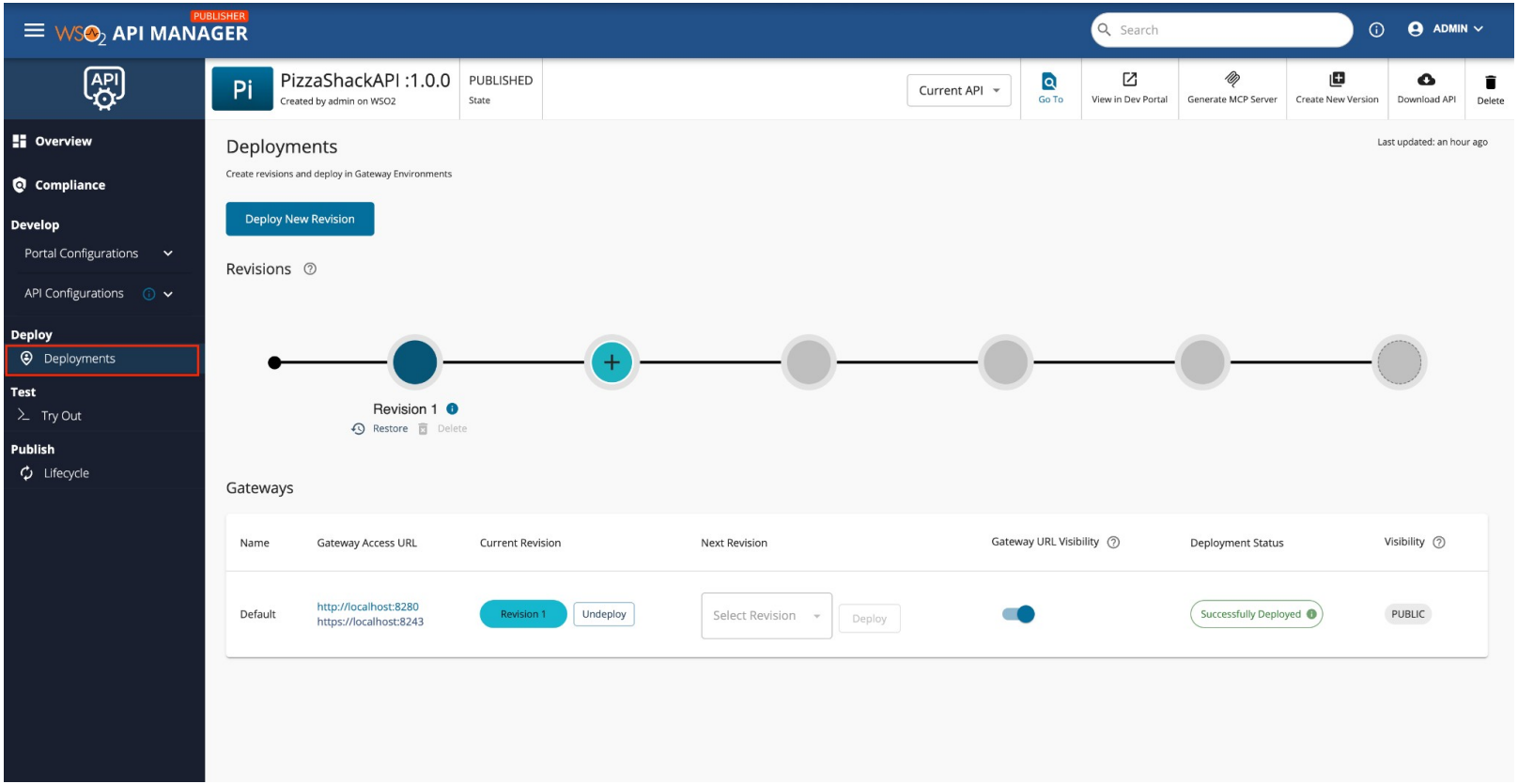
Deselect all business plans to disable subscription validation for the API.



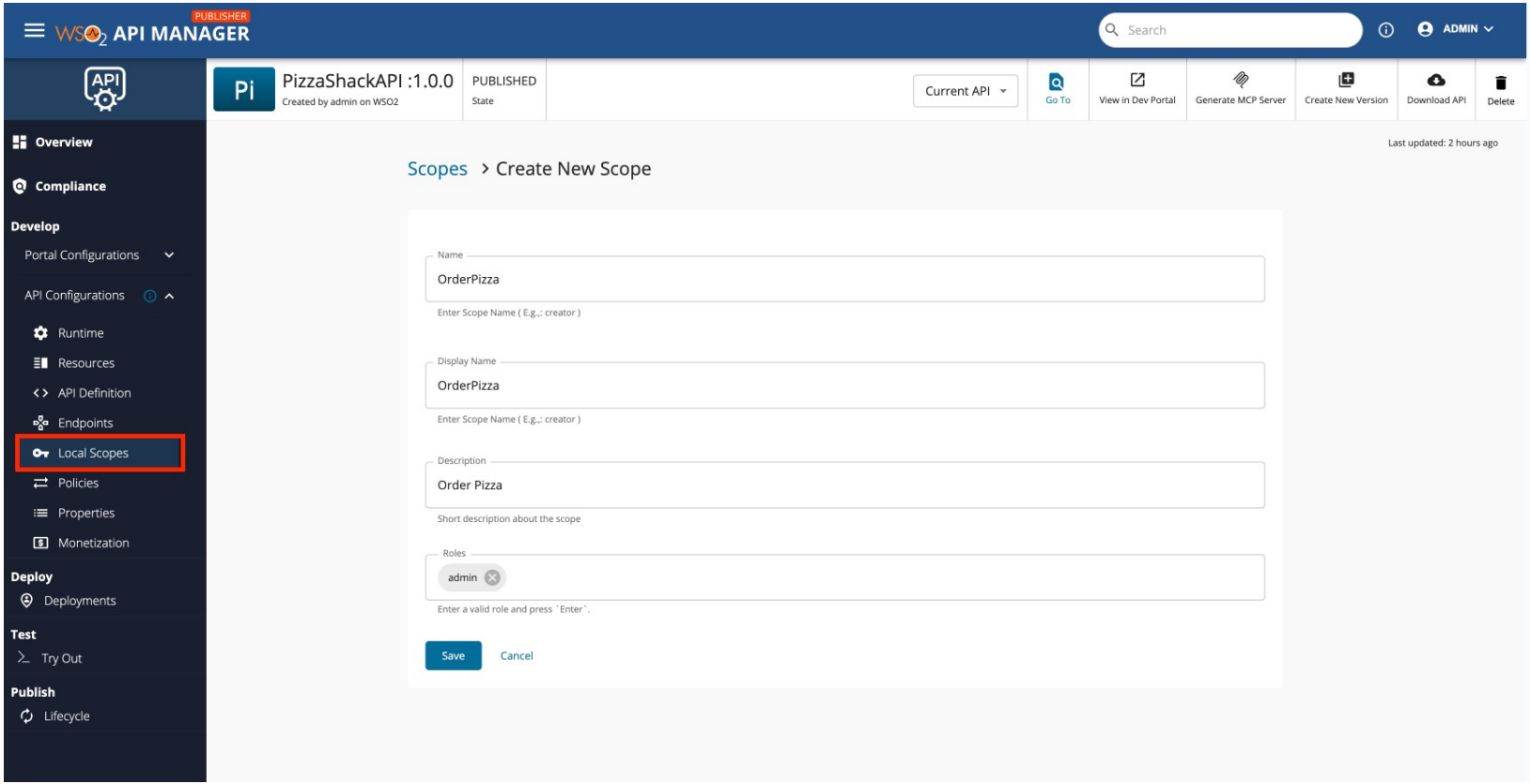
API Definition



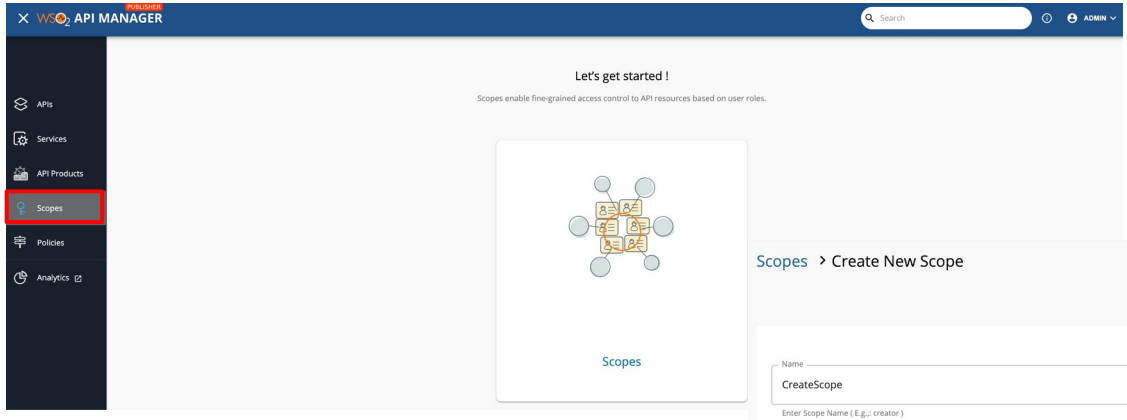
Deployments and API Gateways

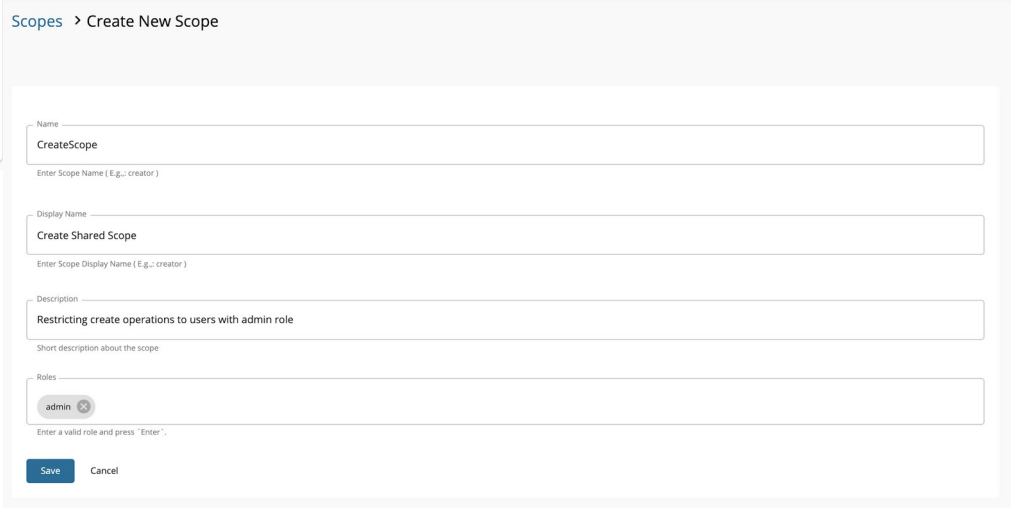


Local Scopes

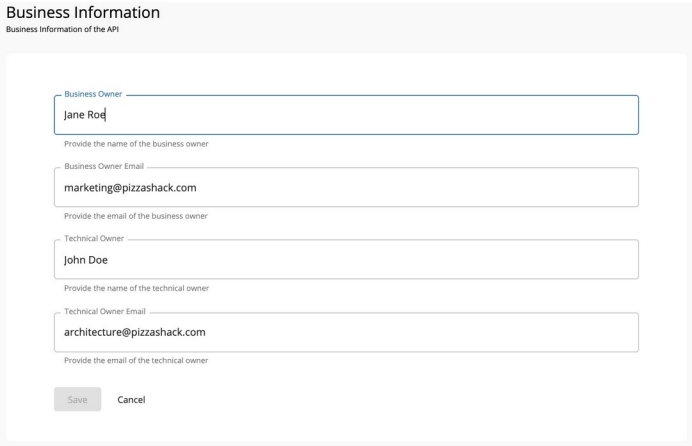


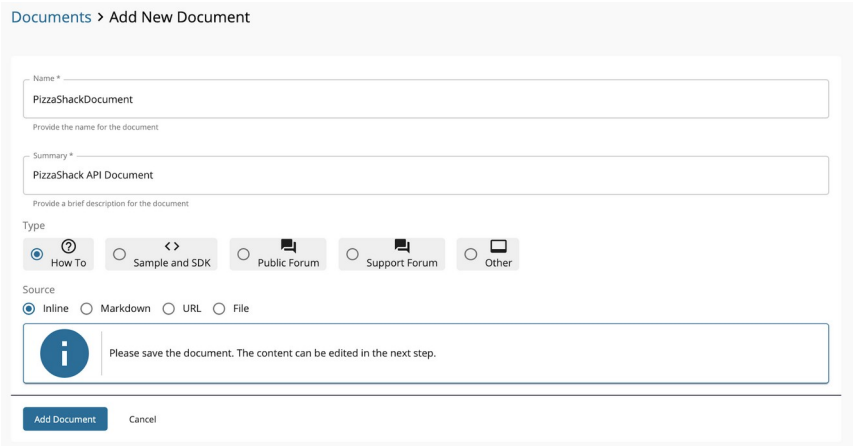
Shared Scopes

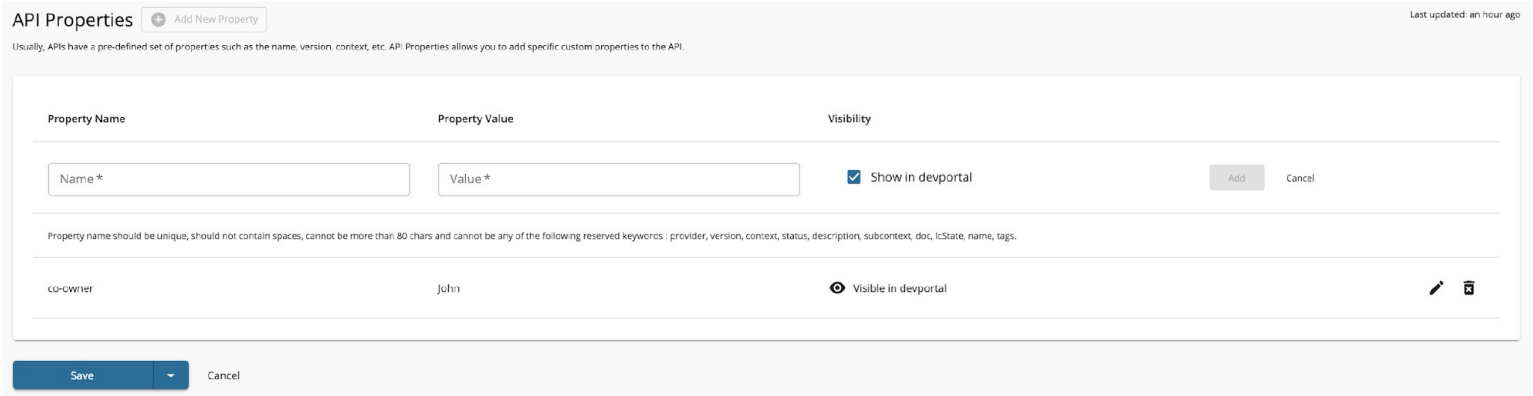




Business Information, API Properties and Documentation



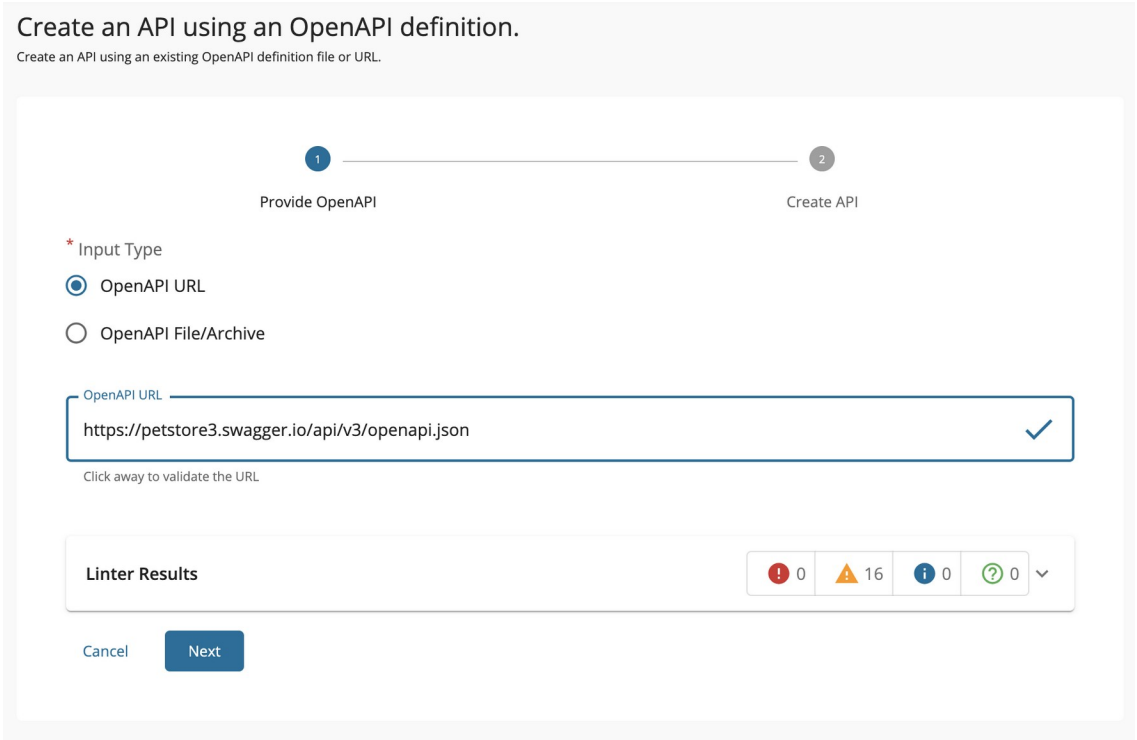




What is Swagger/OpenAPI ?

- An OpenAPI file allows you to describe your entire API, including:
- Available endpoints (/users) and operations on each endpoint (GET /users, POST /users)
- Operation parameters Input and Output for each operation
- Authentication methods
- Contact information, license, terms of use and other information.
- API specifications can be written in YAML or JSON. The format is easy to learn and readable to both humans and machines.

REST API from OpenAPI (Swagger) Definition



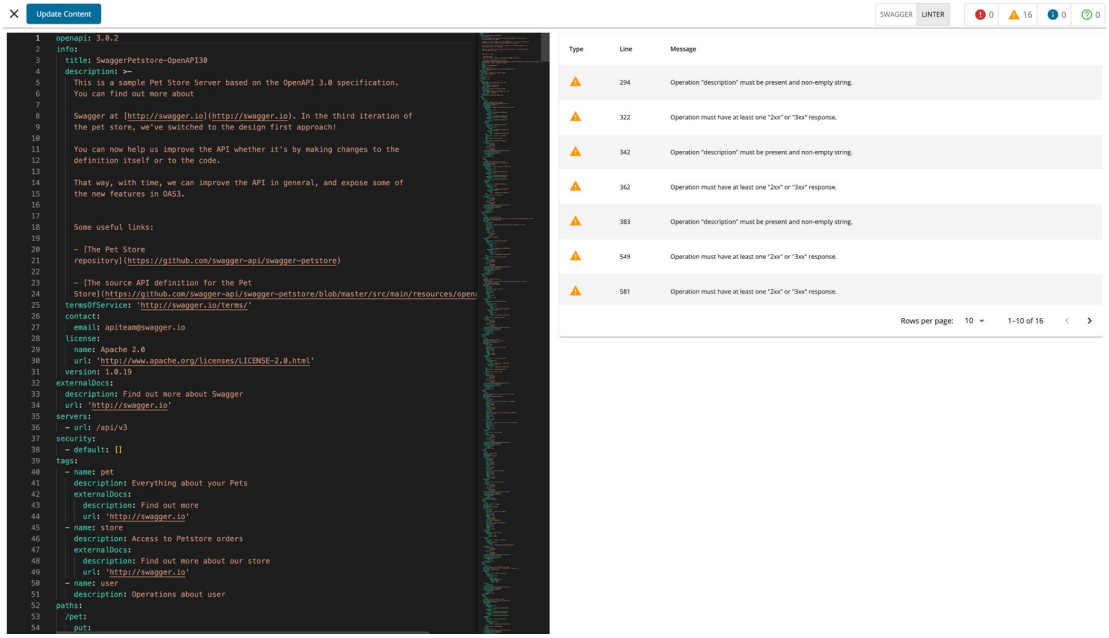
REST API from OpenAPI Definition

Resources Tab



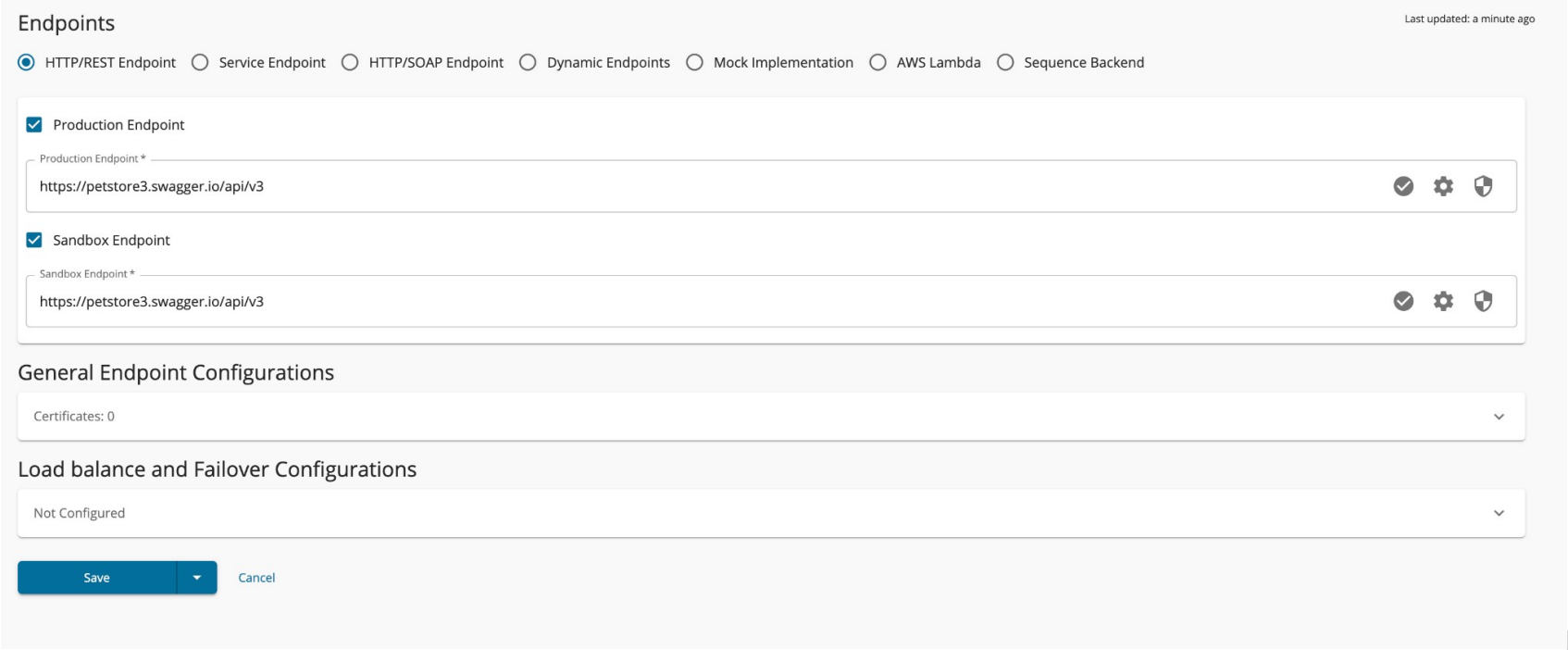
REST API from OpenAPI Definition

Resources can be edited inline



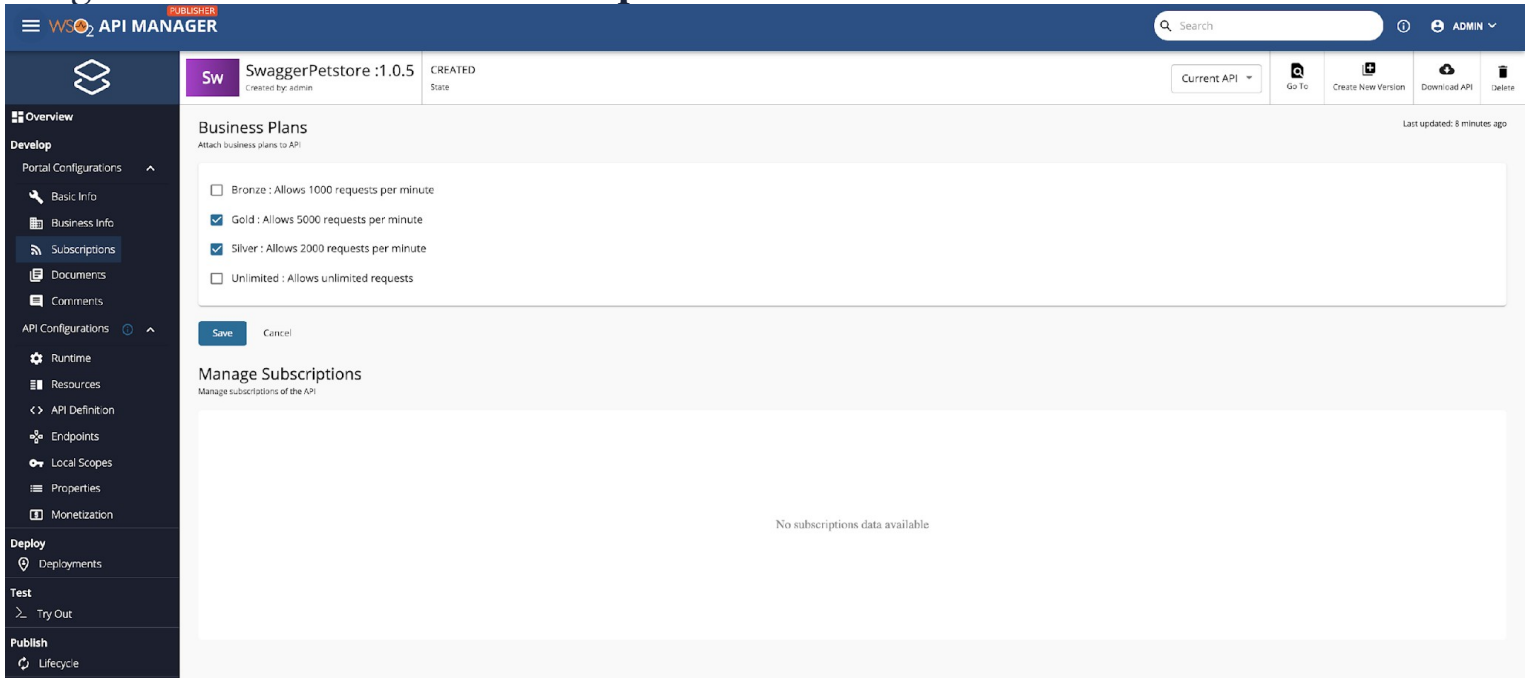
REST API from OpenAPI Definition

Configure **Production** and **Sandbox** endpoints



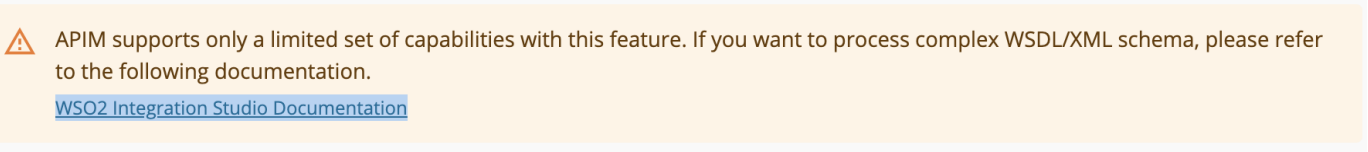
REST API from OpenAPI Definition

Configure Business Plans in **Subscription** Tab

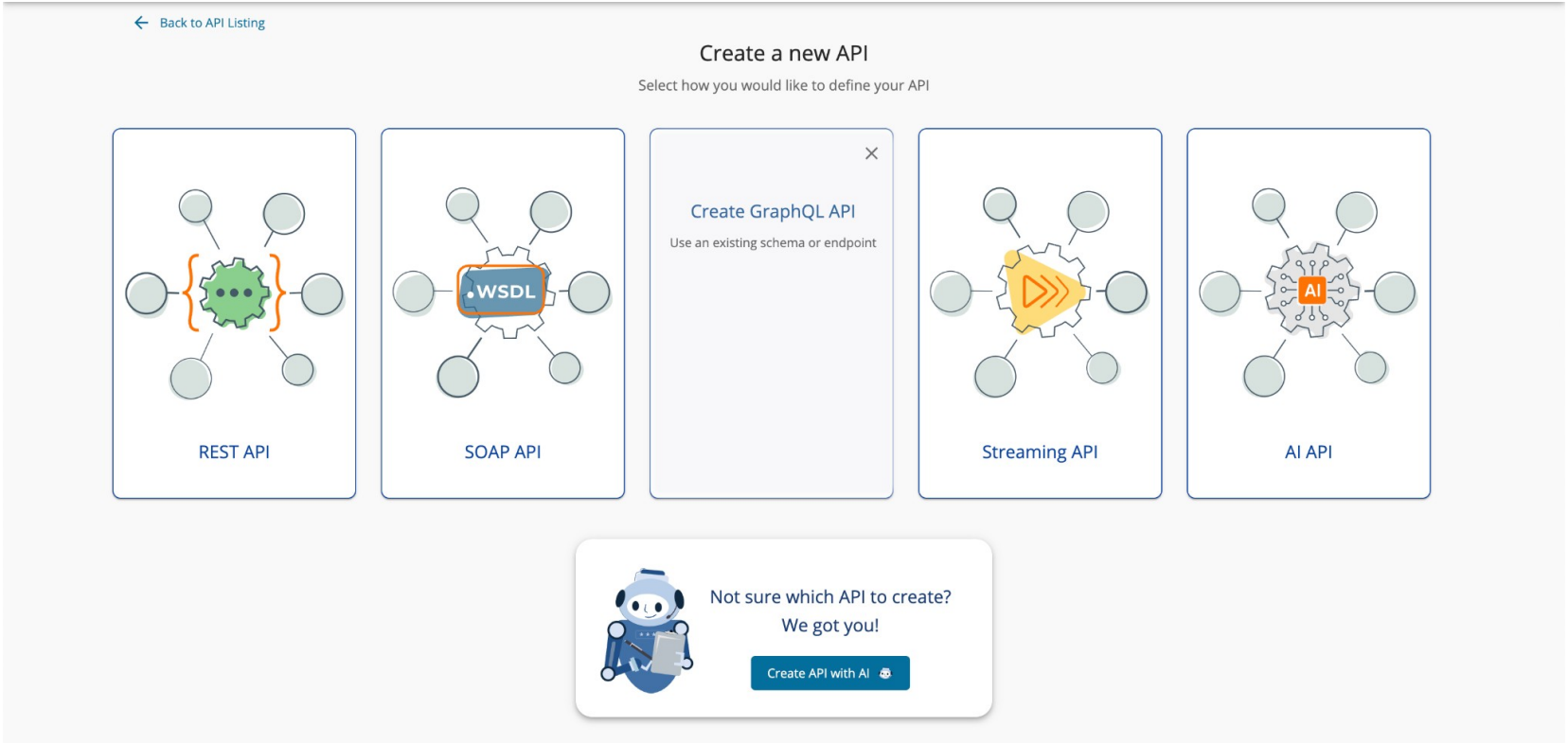


What is SOAP (Simple Object Access Protocol)?

- SOAP is an XML based standard communication protocol that permits processes using different operating systems like Linux and Windows to communicate via HTTP and SOAP APIs are based on this protocol.
- REST API -> Swagger
- SOAP API -> WSDL
- The Swagger specification defines a set of files required to describe REST API.
- WSDL (Web Service Description Language), is an XML based definition language. It is used for describing the functionality of a SOAP based web service.
- APIM supports only a limited set of capabilities for SOAP APIs and it is recommended to use WSO2 Integration studio as an alternative.

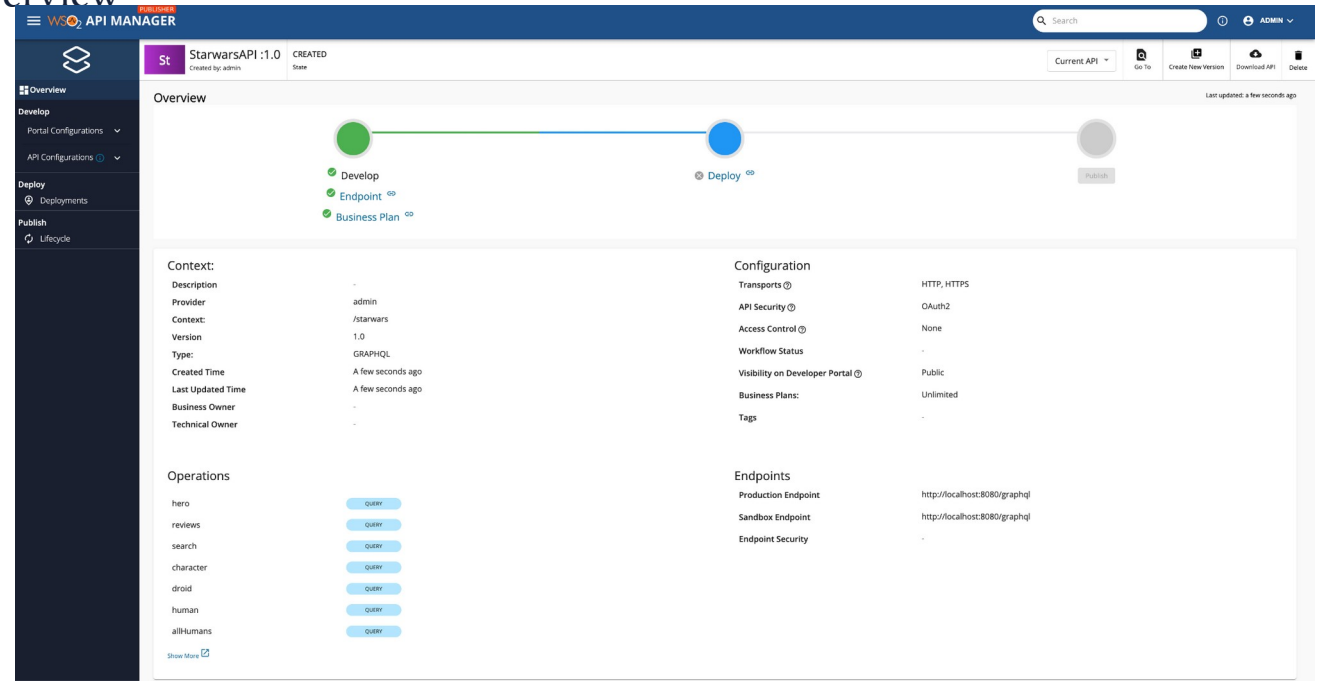


GraphQL API from SDL



GraphQL API from SDL

API Overview



Streaming APIs

- A logical collection of related topics
- Topics represent the different channels
- Topics in Streaming APIs can be compared to the resources in REST APIs.

The actions a **topic** allows you to do are **Subscribe** and/or **Publish**.

- **Subscribe** - A topic of this type allows traffic from the server (backend) to the client.
- **Publish** - A topic of this type allows traffic from the client to the server. The Streaming Integrator component in WSO2 API Manager (WSO2 API-M) supports Streaming APIs via the following main protocols, which are compatible with HTTP.
    - WebSocket
    - WebSub (WebHook)
    - Server Sent Events (SSE)

WebSocket API

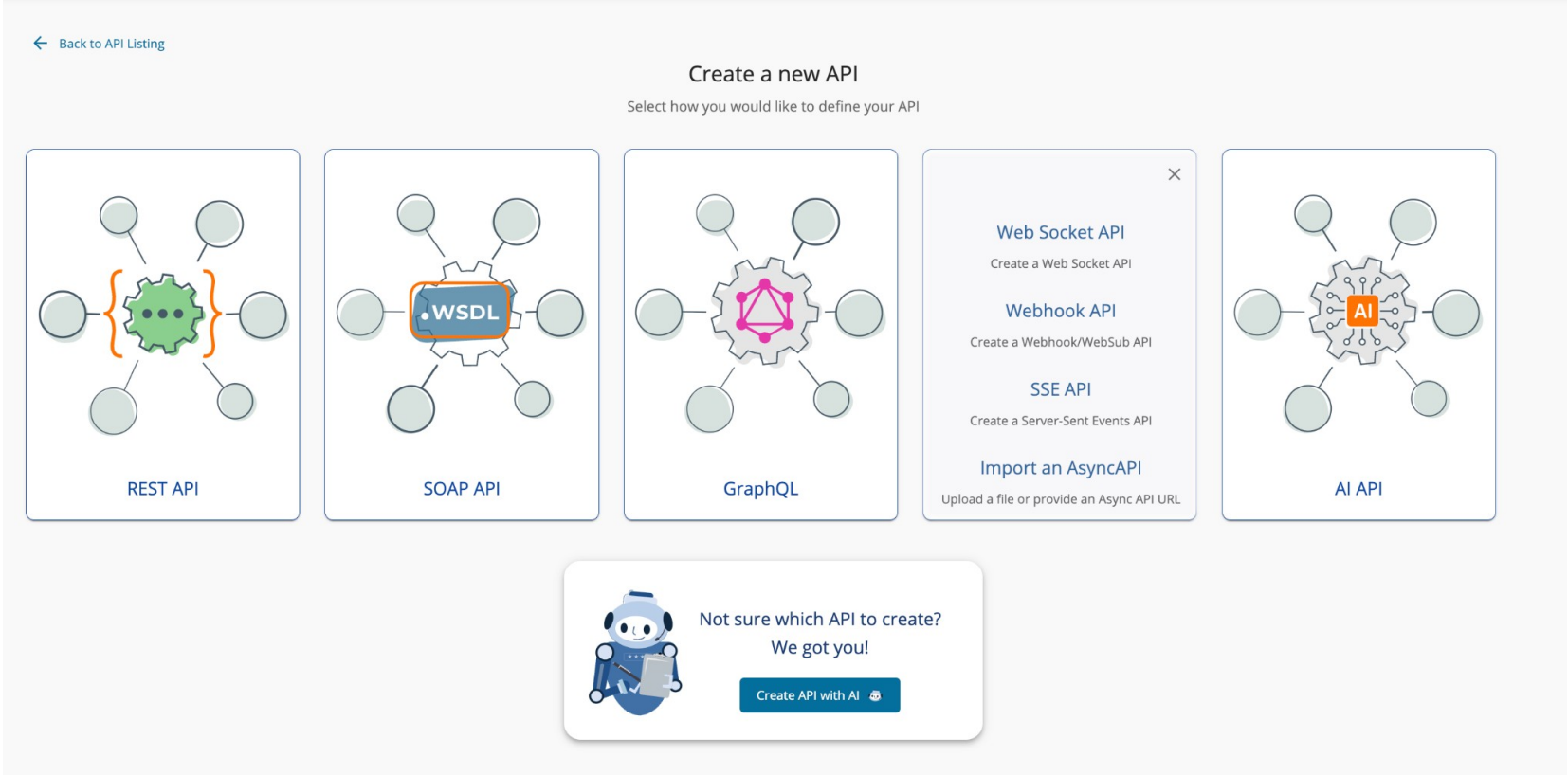
- Protocol similar to HTTP that is part of the HTML5 specification.
- Full-duplex communication between the client and the server over a single connection.
- Advantages

  Reduce unnecessary network traffic and latency.

  Allow streaming through proxies and firewalls while simultaneously supporting upstream and downstream communication.

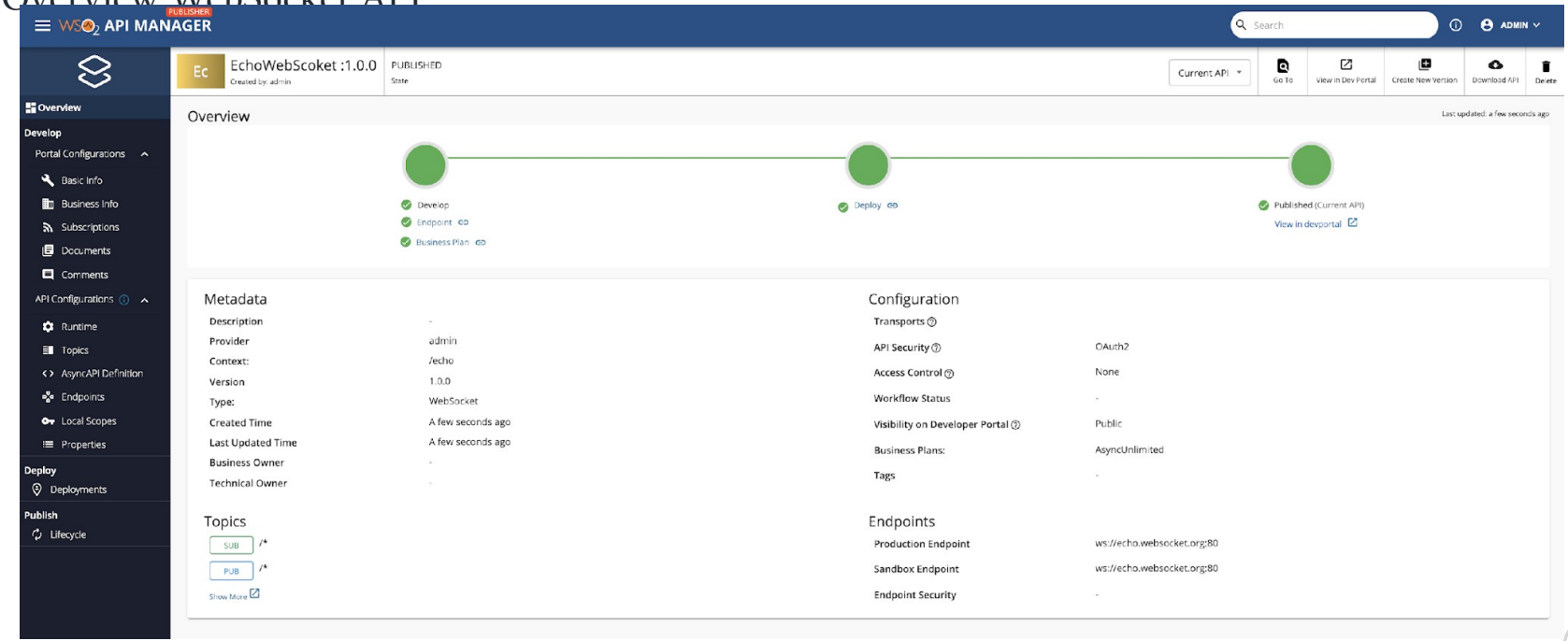
  Be backward compatible with the pre-WebSocket world by starting up as an HTTP connection before switching to WebSocket frames.

WebSocket API

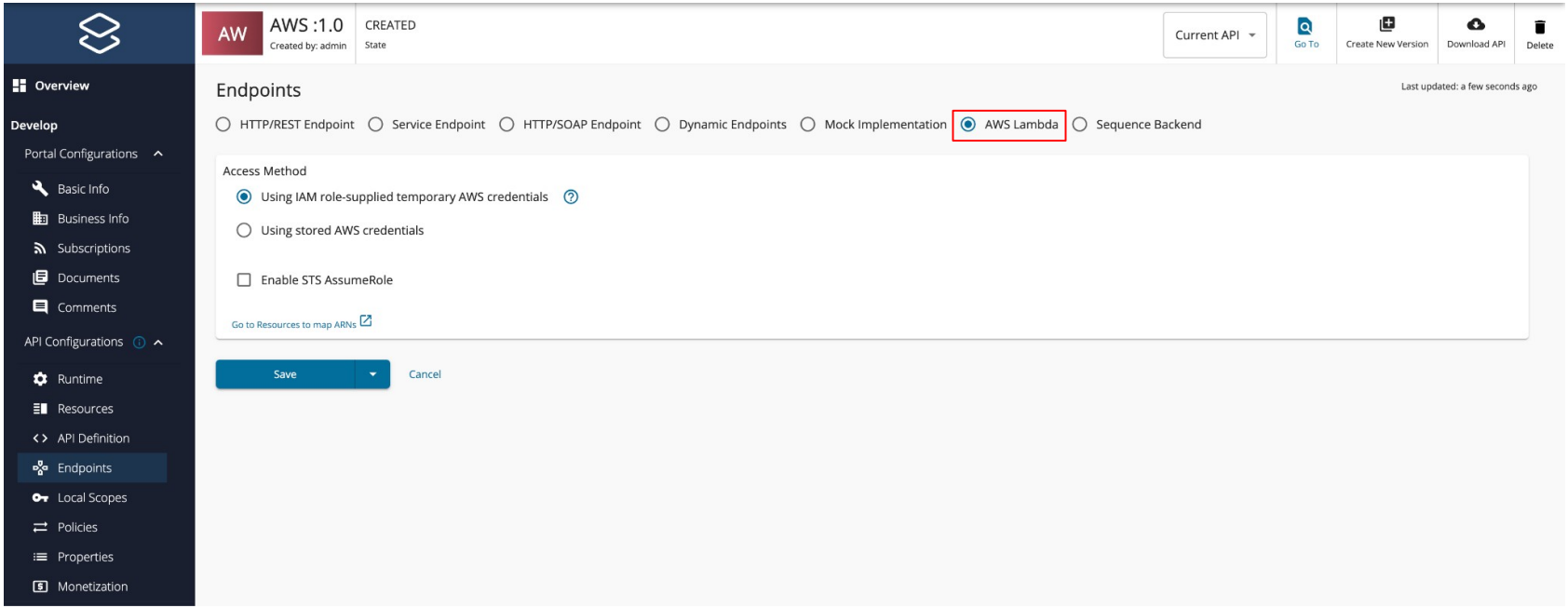


WebSocket API

Overview WebSocket API

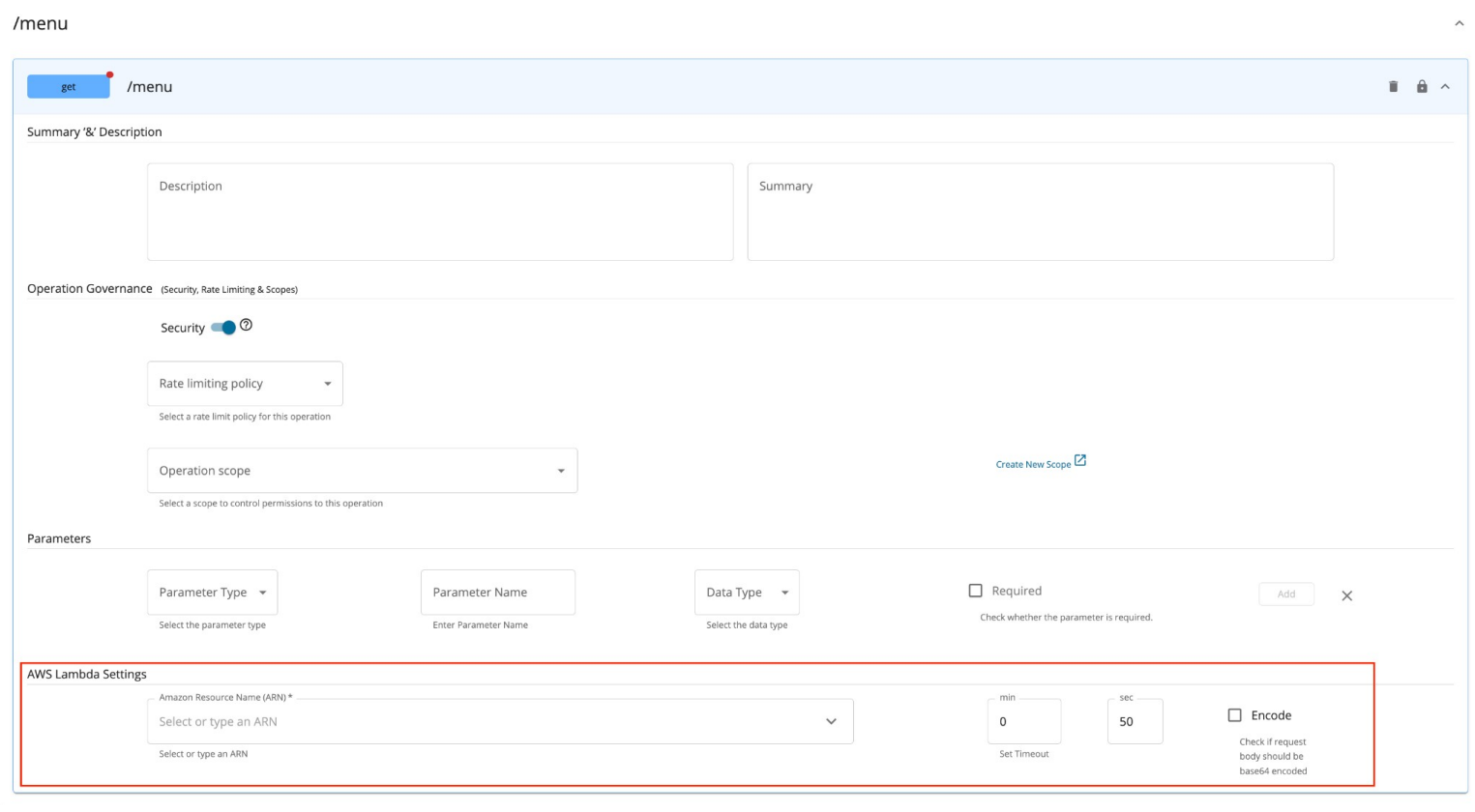


API with AWS Lambda Endpoint



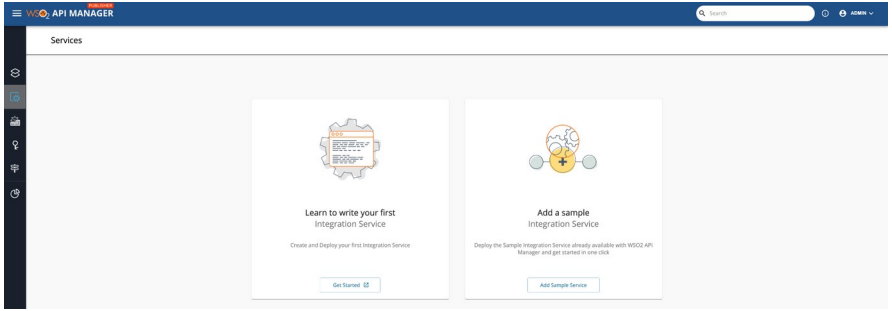
Doc Link: Create and Publish an AWS Lambda API

AWS Lambda - Select Resource Level ARNs



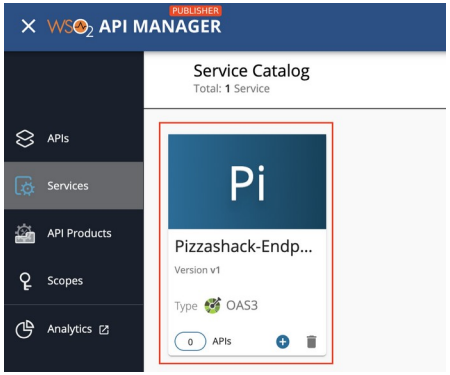
Create API Using a Service

- The service catalog in WSO2 API Manager contains services that correspond to either the Micro Integrator or the Streaming Integrator.
- The Micro Integrator services correspond to the REST API backend services
- The Streaming Integrator services correspond to the Streaming API (i.e., WebSocket, WebHook/WebSub or SSE) backend services that are managed by the respective integration layer (Micro Integrator or Streaming Integrator).

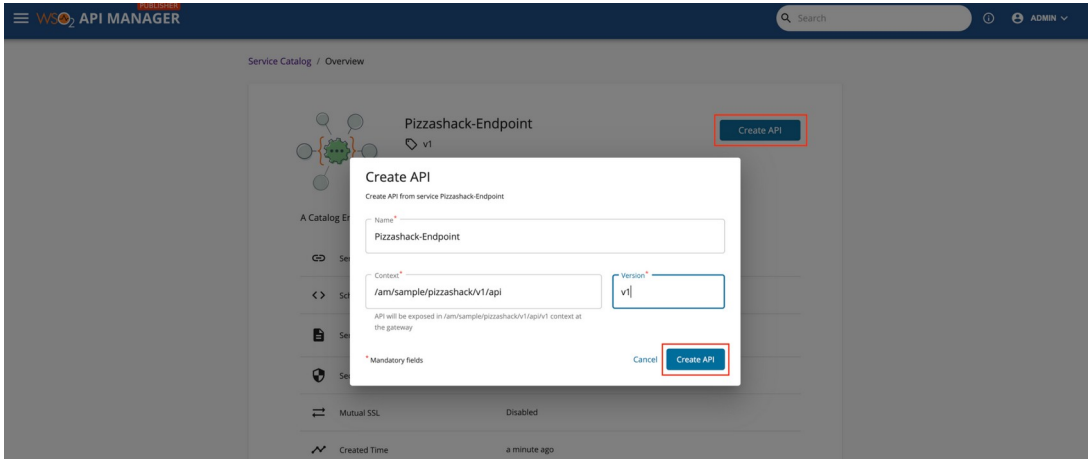


Create an API Using a Service

1. Discover the service

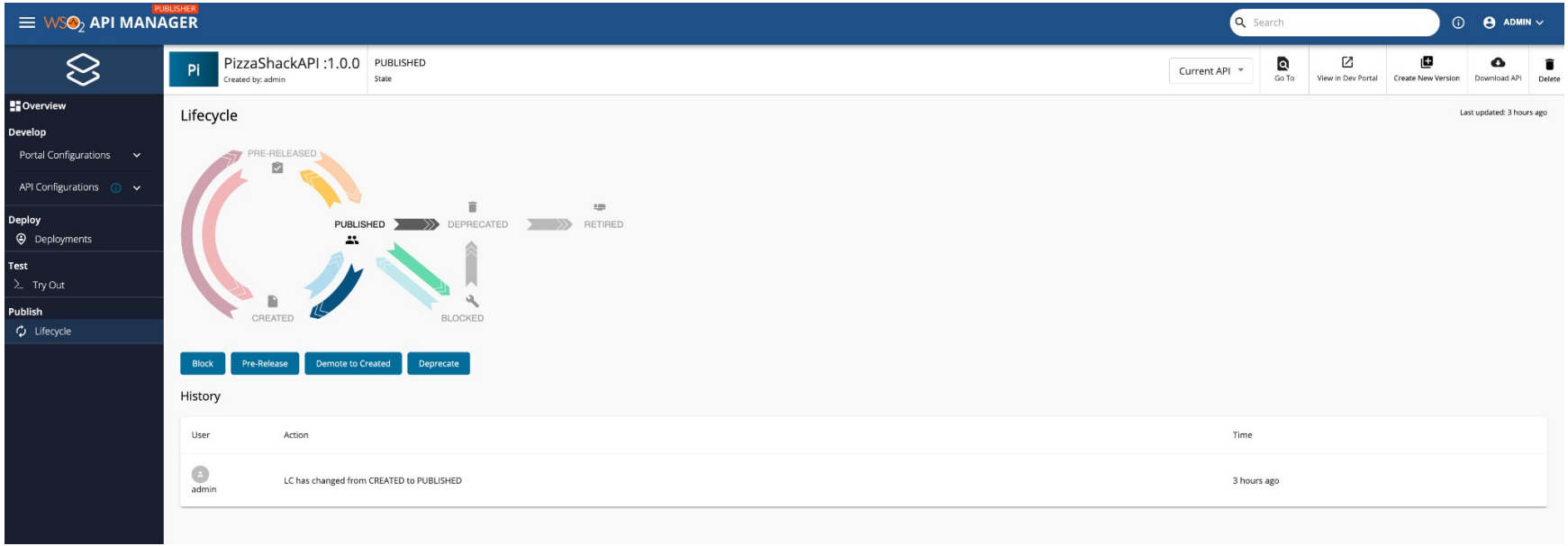


2. Create API from the Service



Create an API from a Service

API Lifecycle



Managing APIs

WSO2

Managing APIs

- Changing the Lifecycle State of APIs
- Deploy APIs
- API Versioning
- Managing API Policies

API Lifecycle

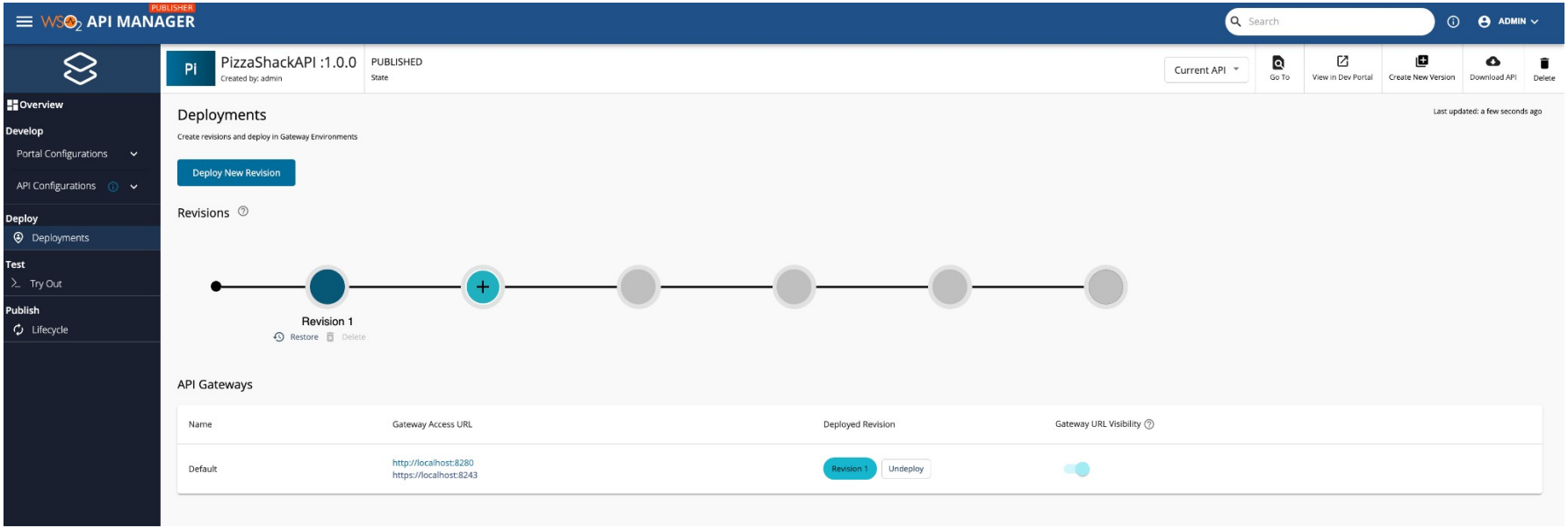
States which represent the stages that an API has in the process of starting to develop an API until its retirement.

- Created
- Pre-Released
- Published
- Blocked
- Deprecated
- Retired

Docs Link: API Lifecycle

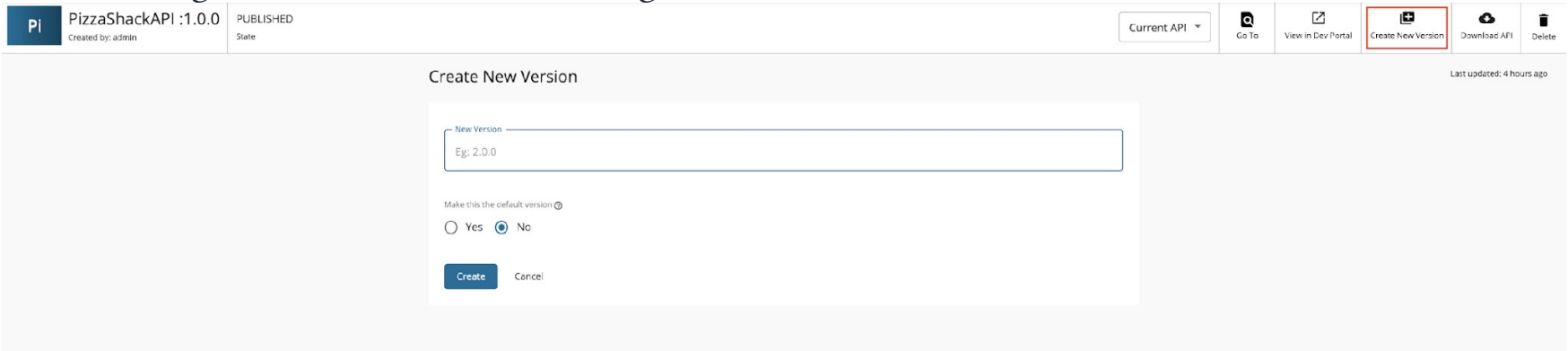
Deploying APIs

API Deploying is the process of making the API available for invocation.

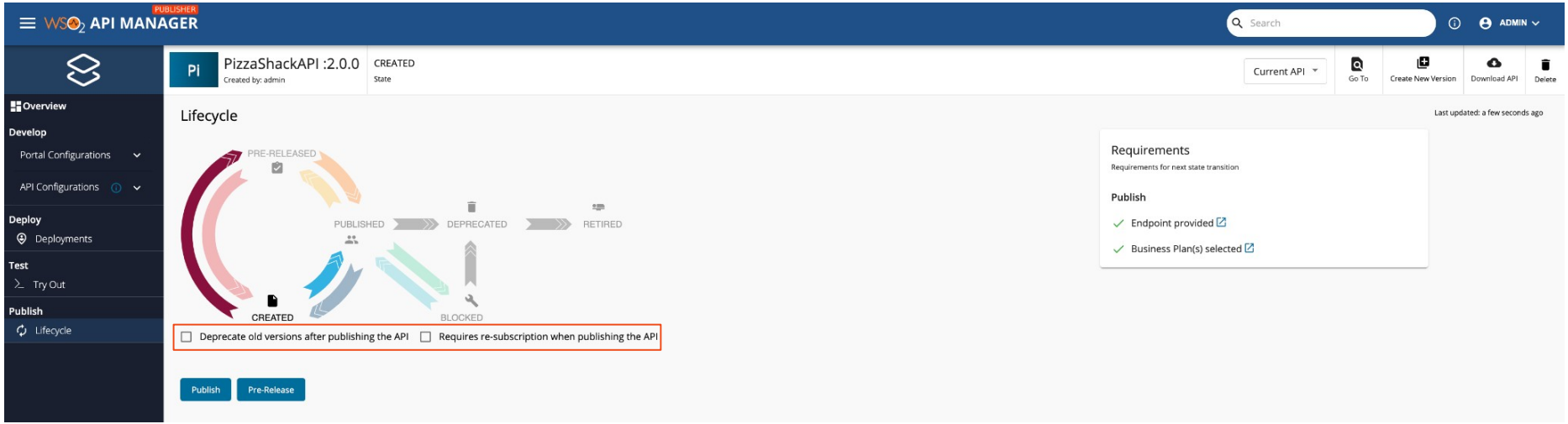


API Versioning

Creating a new version of an existing API.



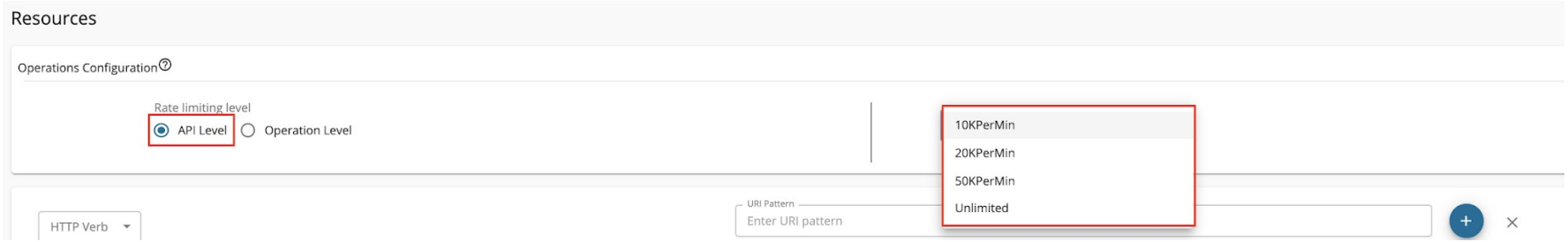
API Versioning



Managing API Policies

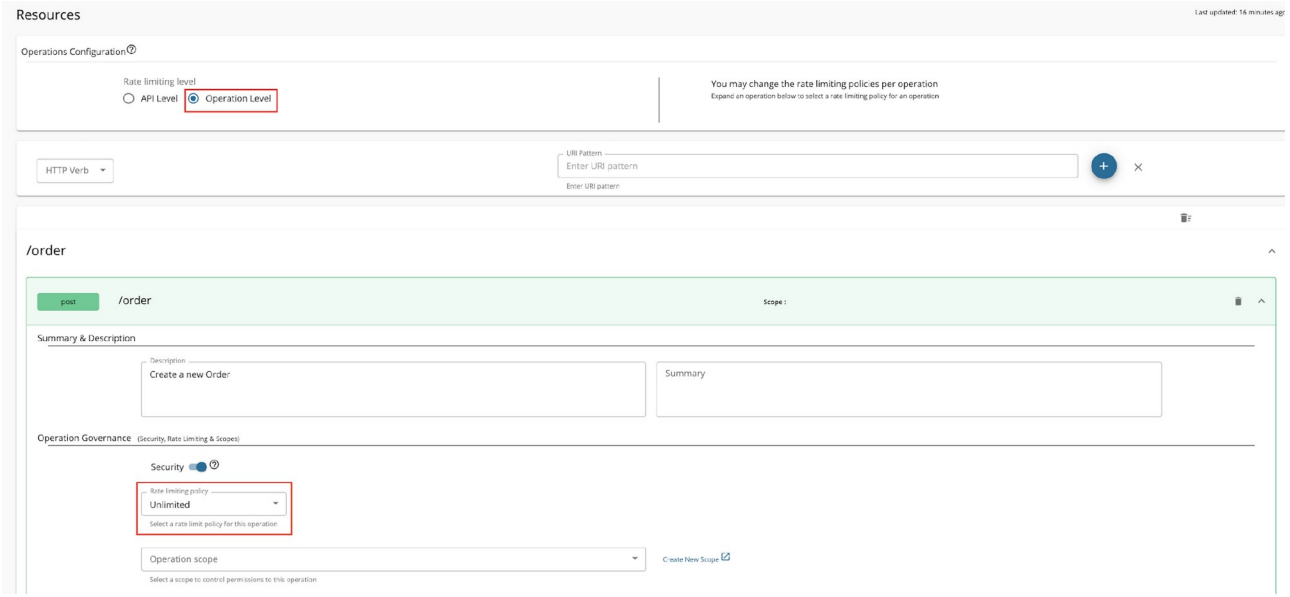
Enforce API Level/ resource level usage policies (Rate limiting policies) to the API.

API Level Policies



Managing API Policies

Operation Level Policies

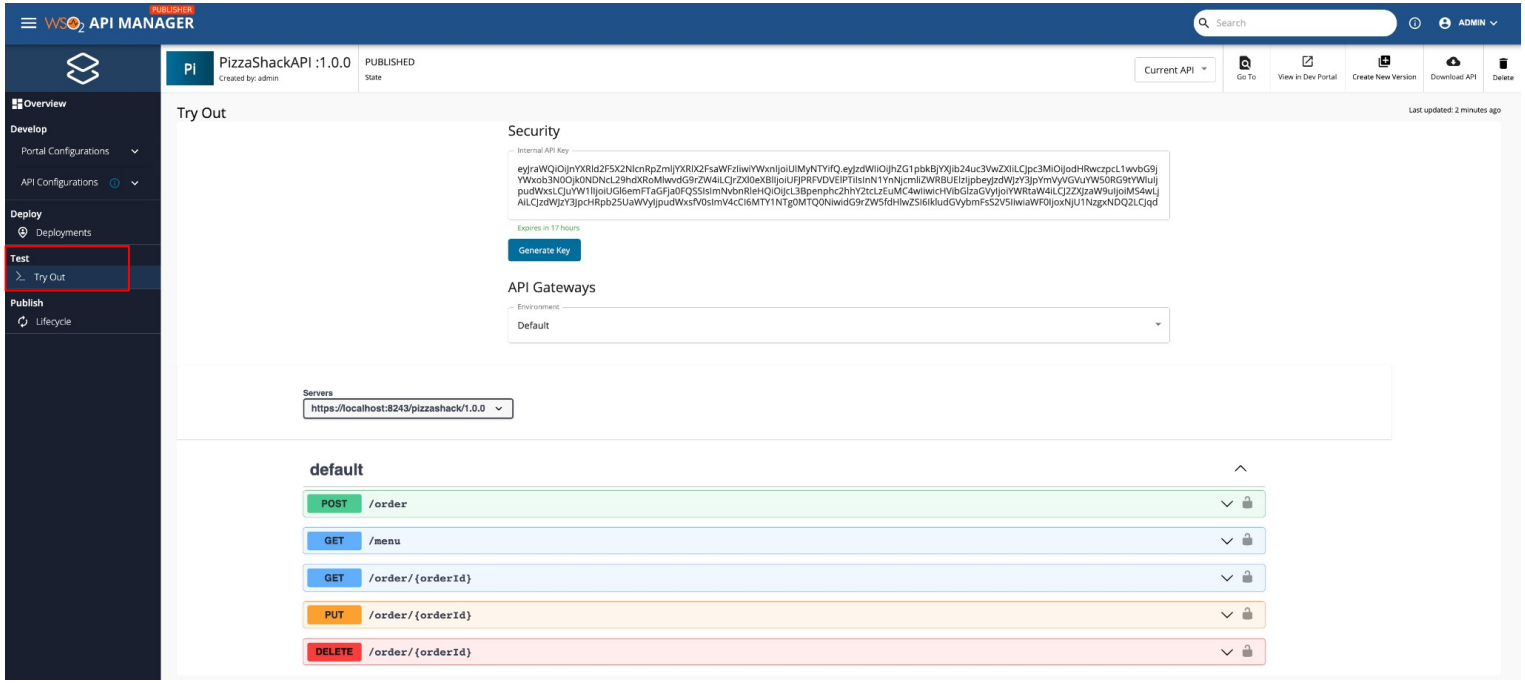


Testing APIs

WSO2

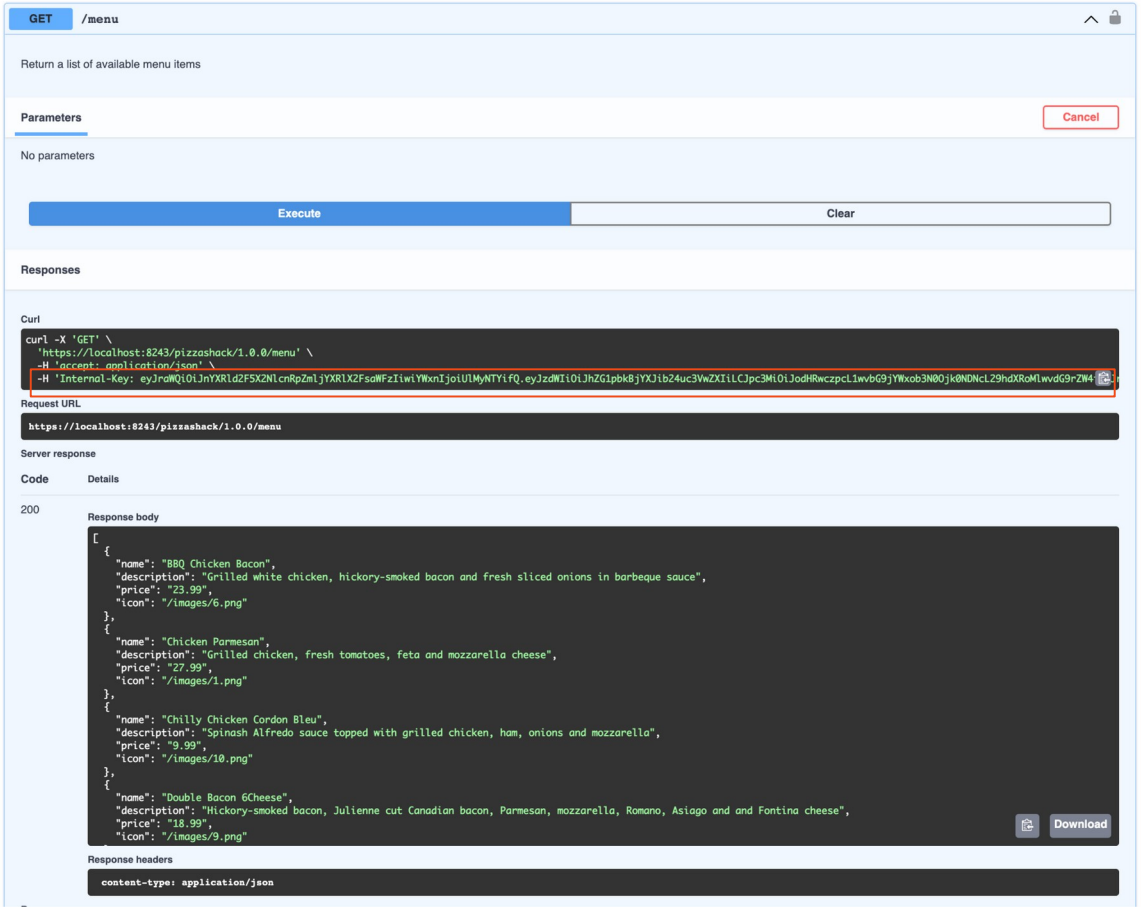
Testing APIs

Testing APIs from the Publisher Portal (API has to be deployed first)



Testing APIs

Test the resource

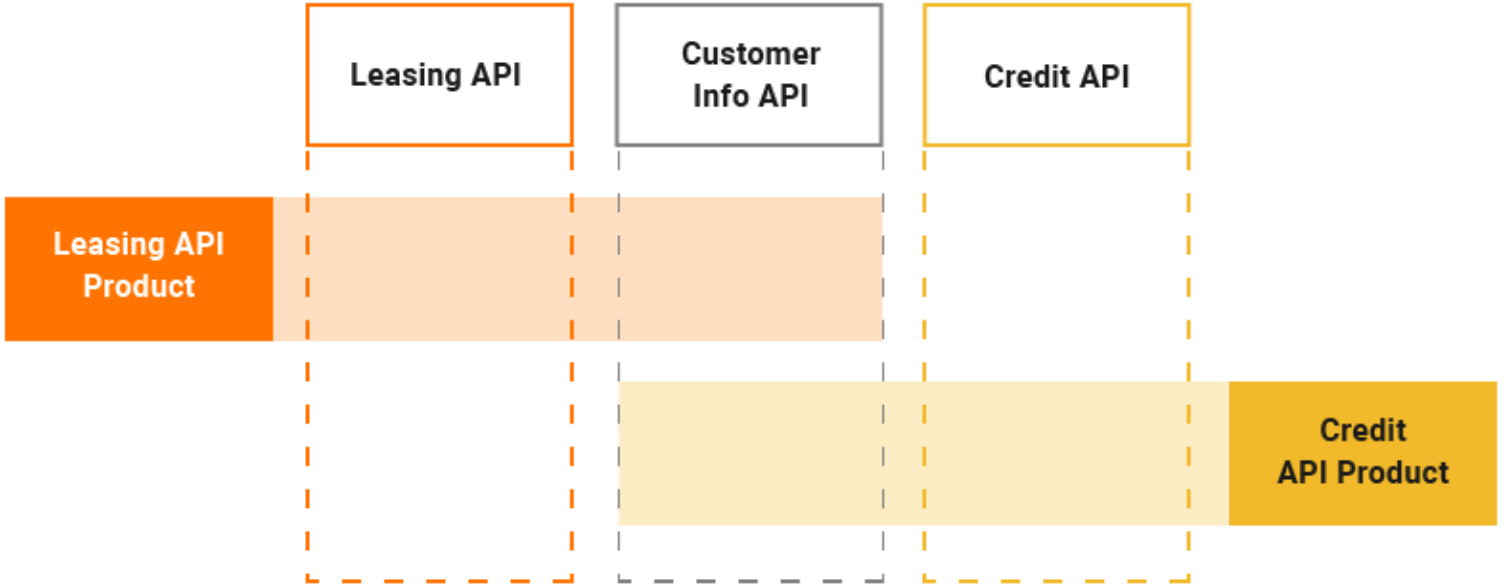


API Products

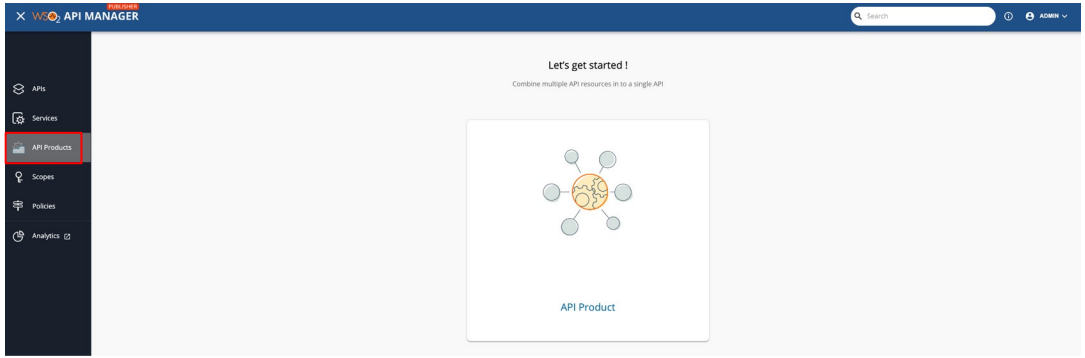
WSO2

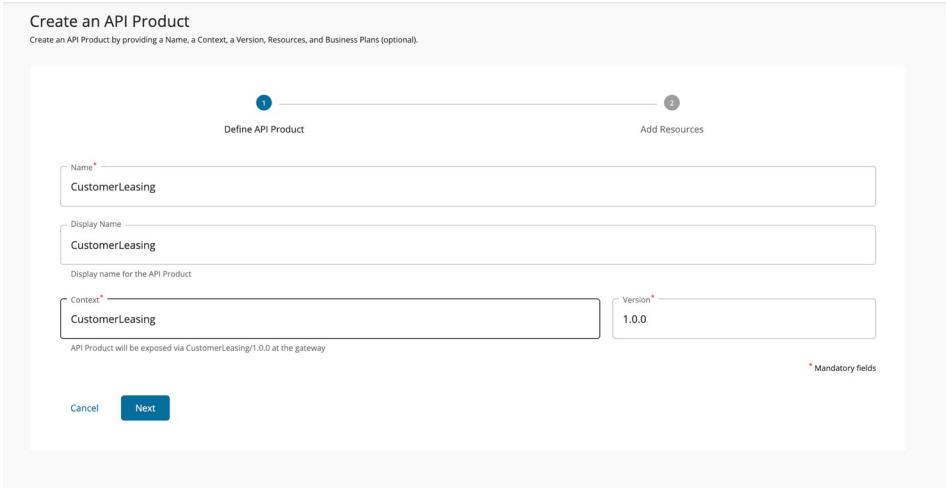
API Product

Packaging mechanism that you can use when you need to bundle a preferred set of resources from multiple APIs and expose it as a separate API interface.



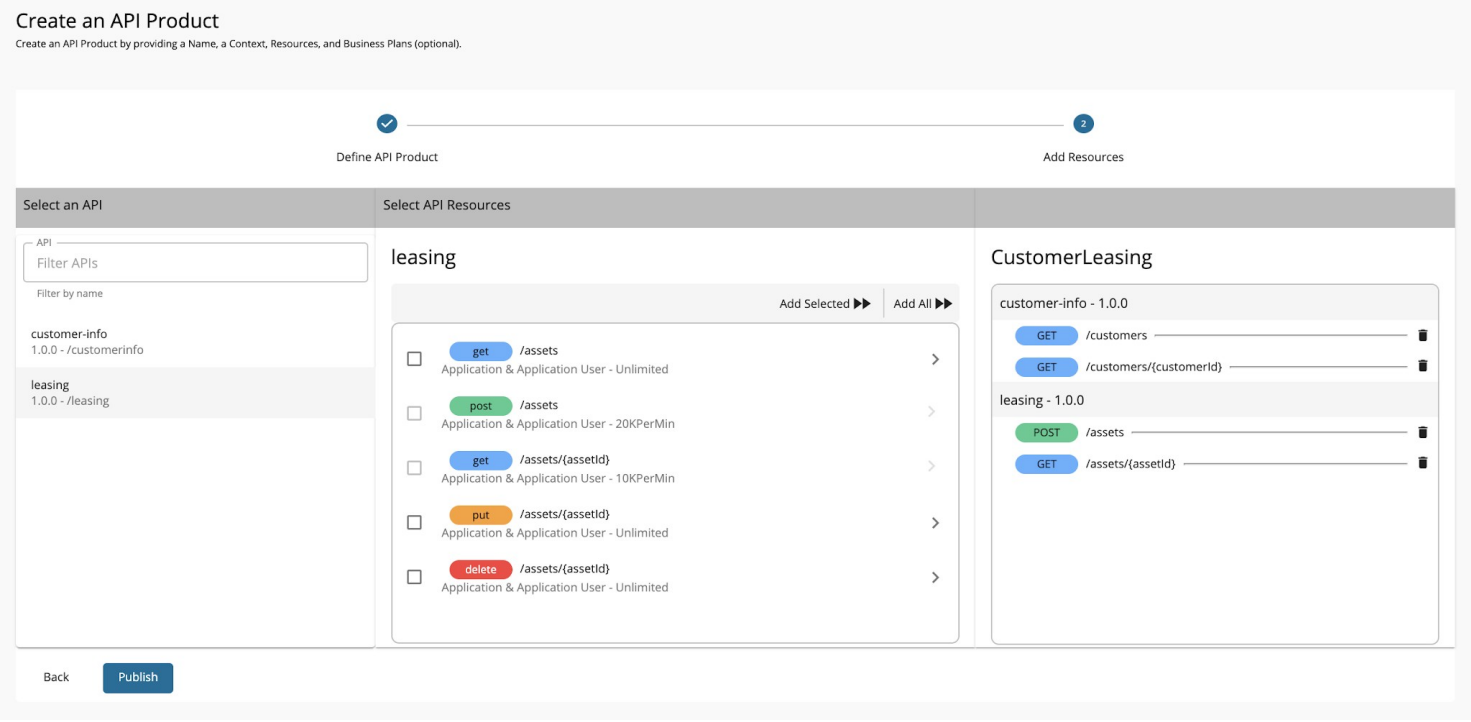
Creating an API Product





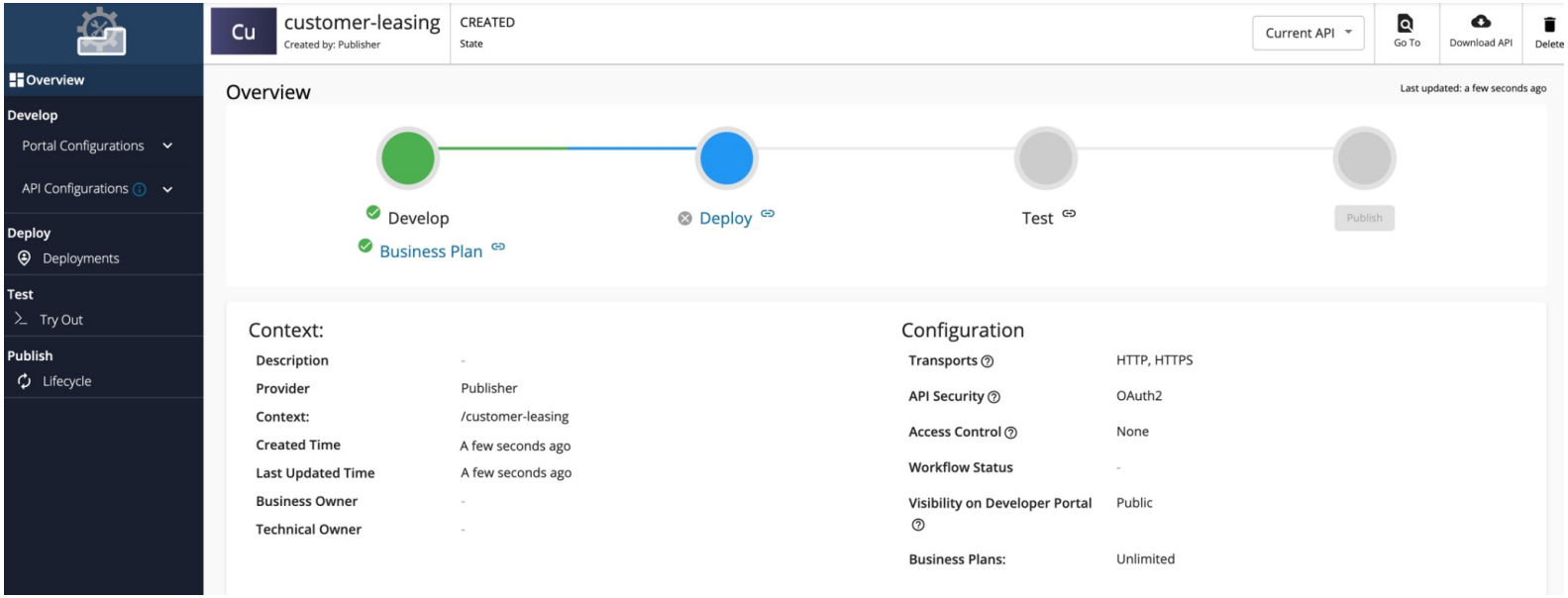
Publishing an API Product

Add Resources from APIs



Publishing an API Product

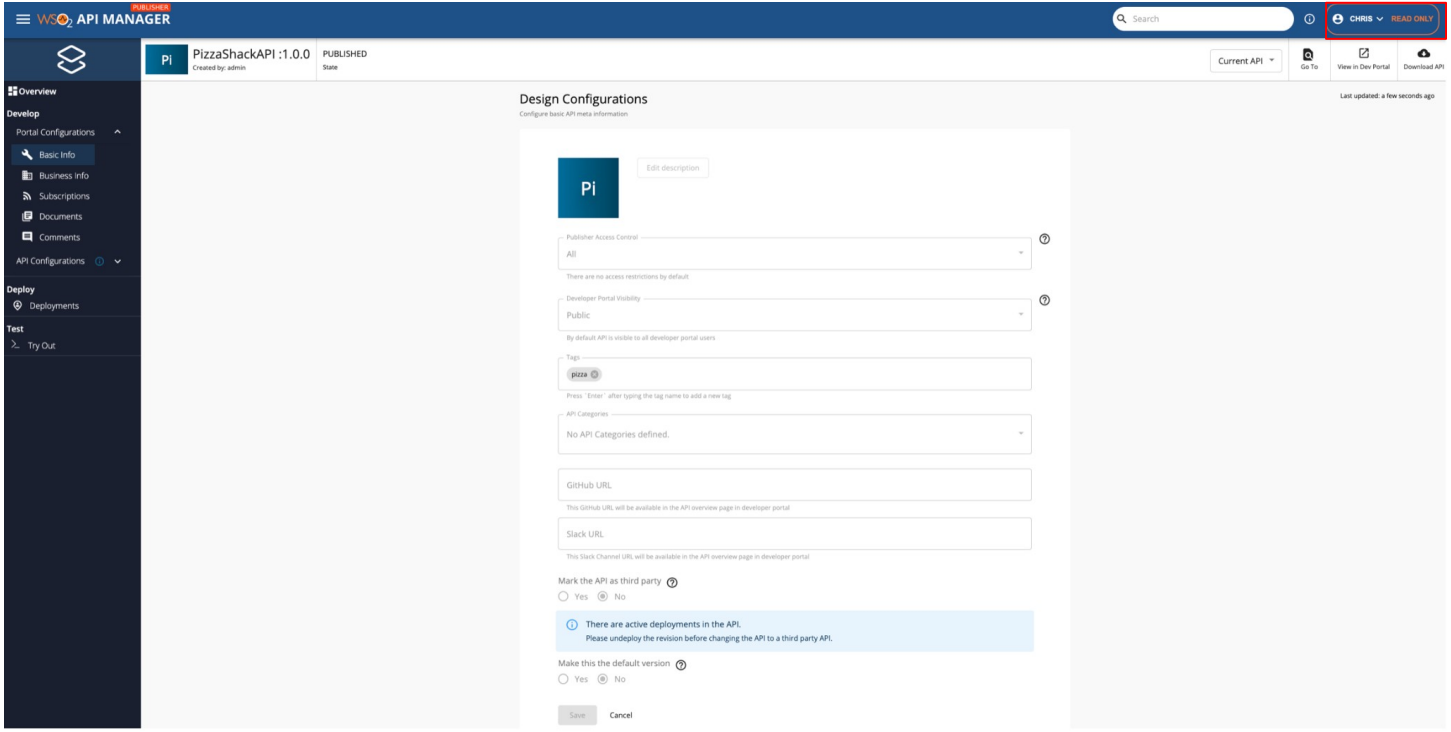
Created API Product Overview



Publisher Portal in Read Only Mode

- Users who have the view/read only permissions can only view the API, API Product and Service details in the Publisher Portal.
- The read only user should have the following scopes - apim:api_view and apim:publisher_settings.
- WSO2 API-M provides a pre-defined role named Internal/observer, which is used to group all the read-only users.

Publisher Portal in Read-Only Mode



Let’s try it out!

Creating and Publishing an API Product through the Publisher

WSO2